

BFS Traversal

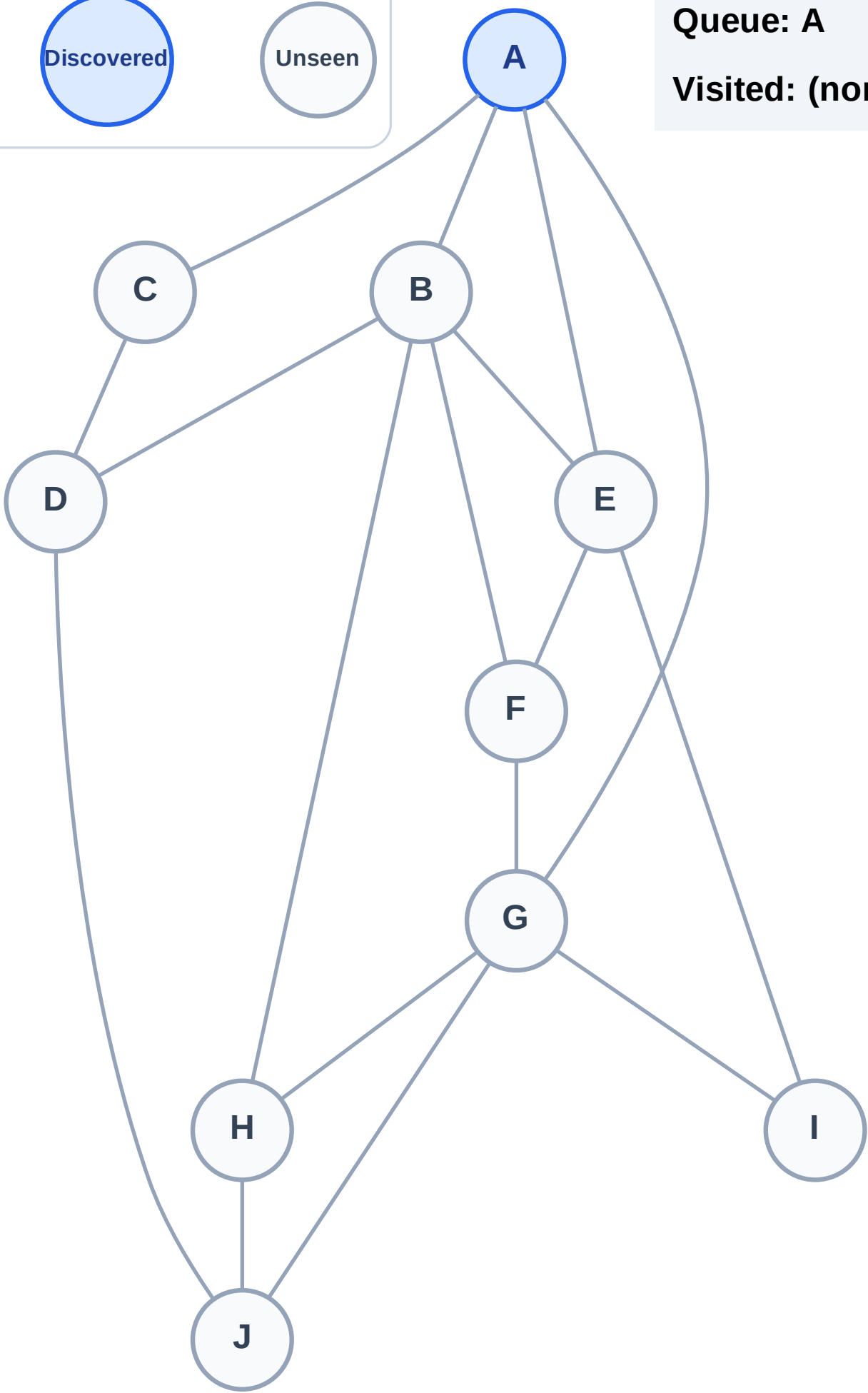
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

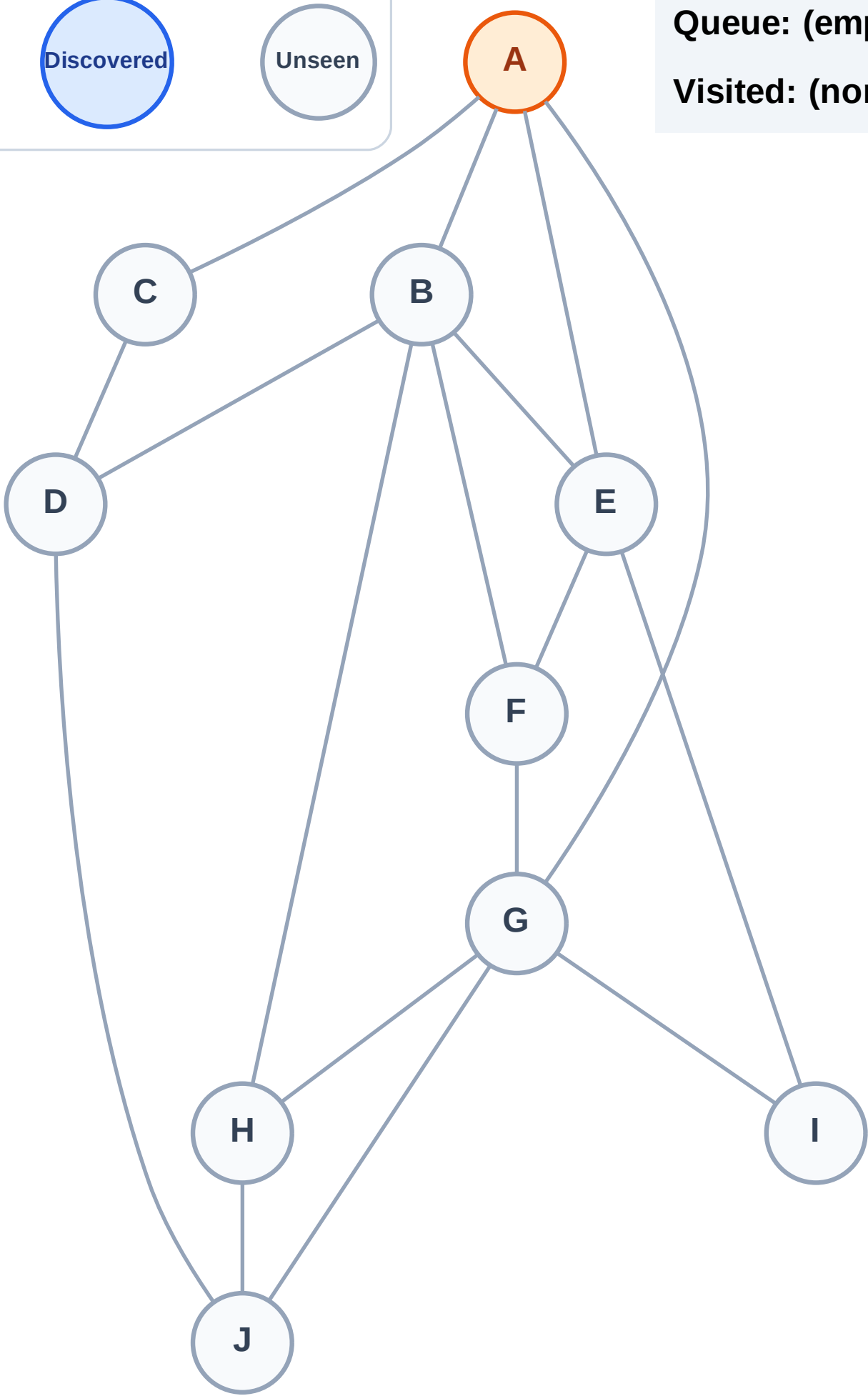
Step 2: Process node A

Legend



Queue: (empty)

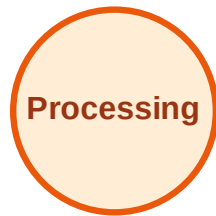
Visited: (none)



BFS Traversal

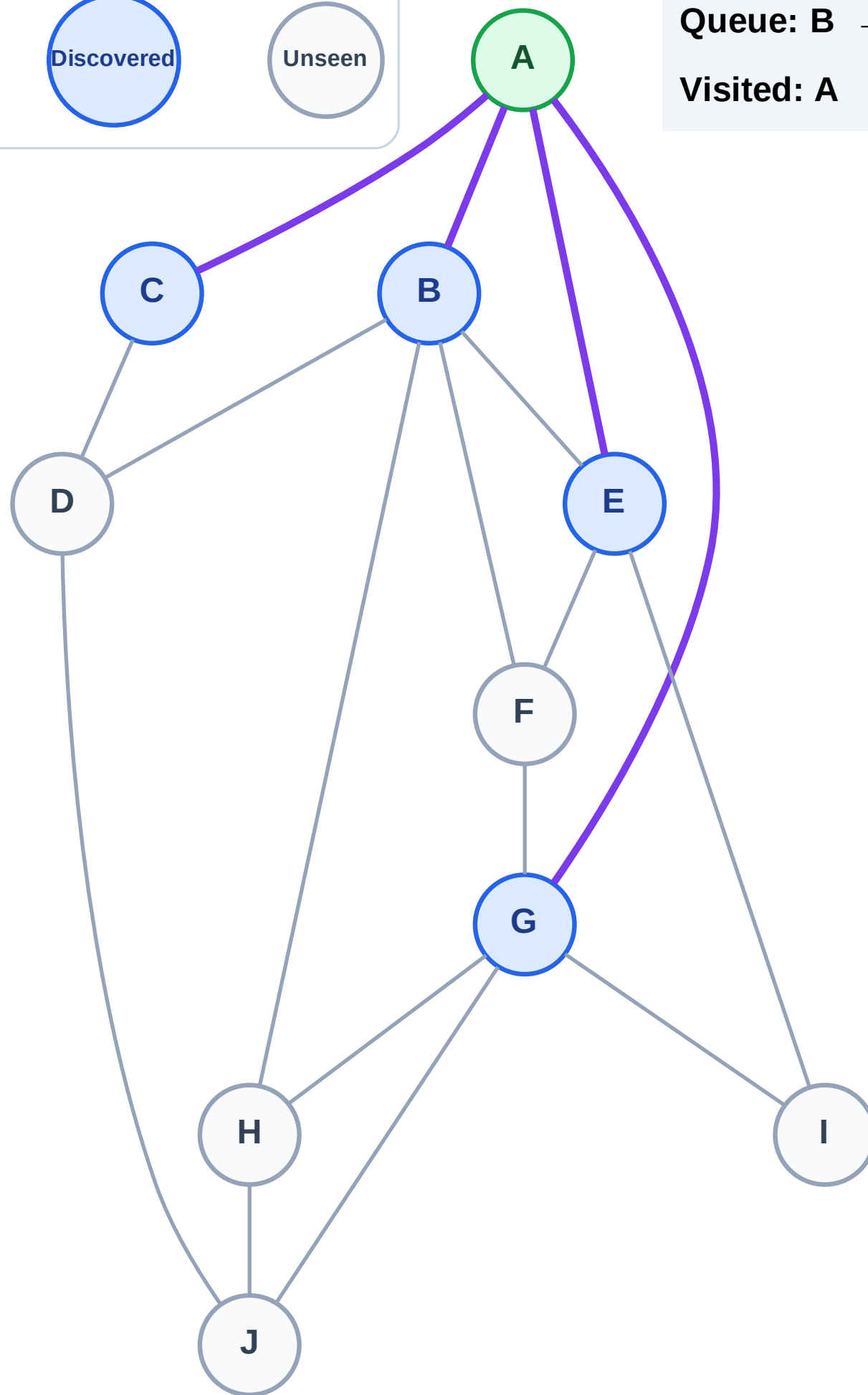
Step 3: Discover B, C, E, G from A

Legend



Queue: B → C → E → G

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

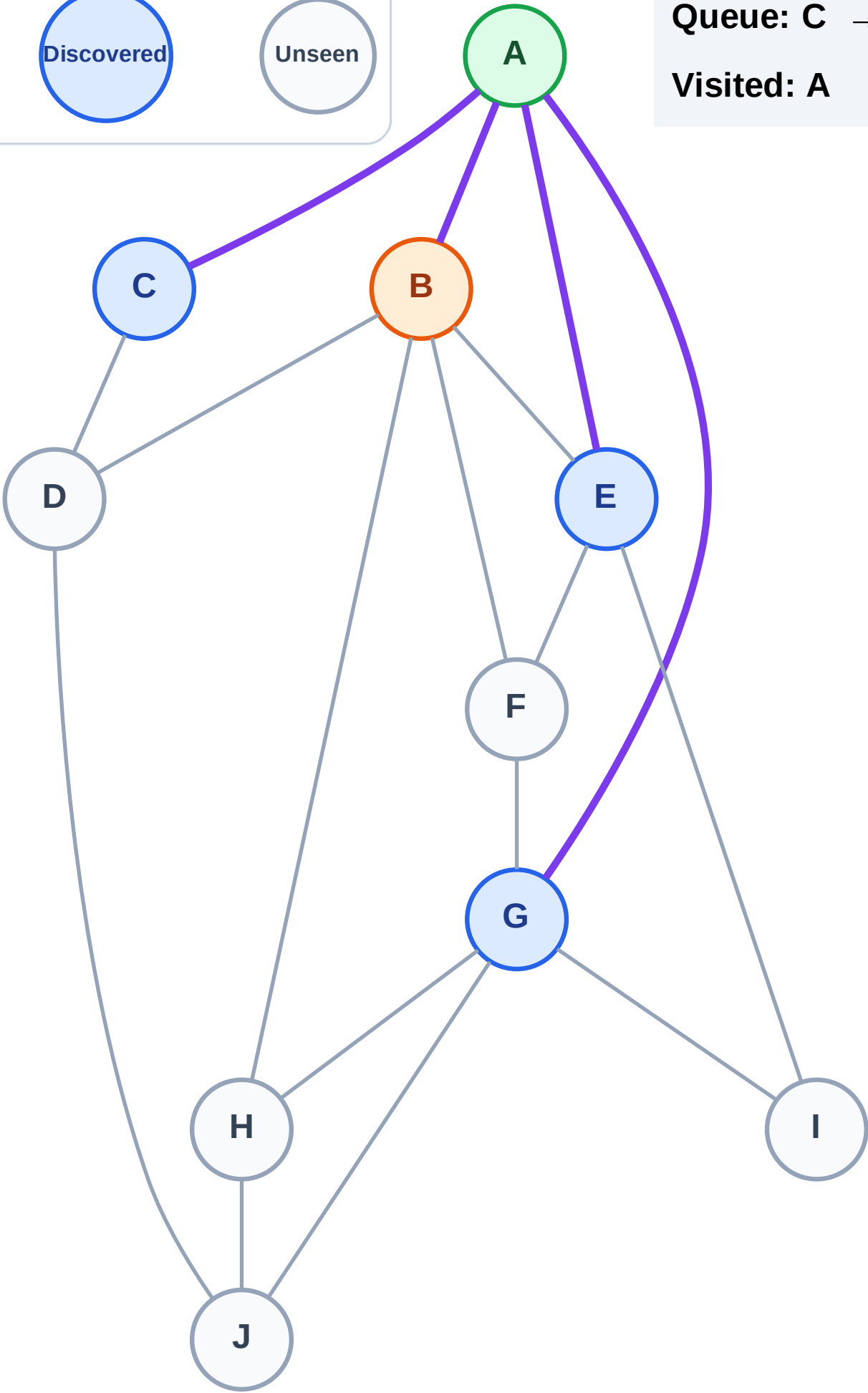
Processing

Discovered

Unseen

Queue: C → E → G

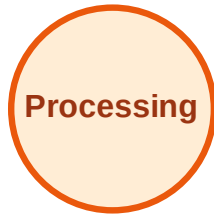
Visited: A



BFS Traversal

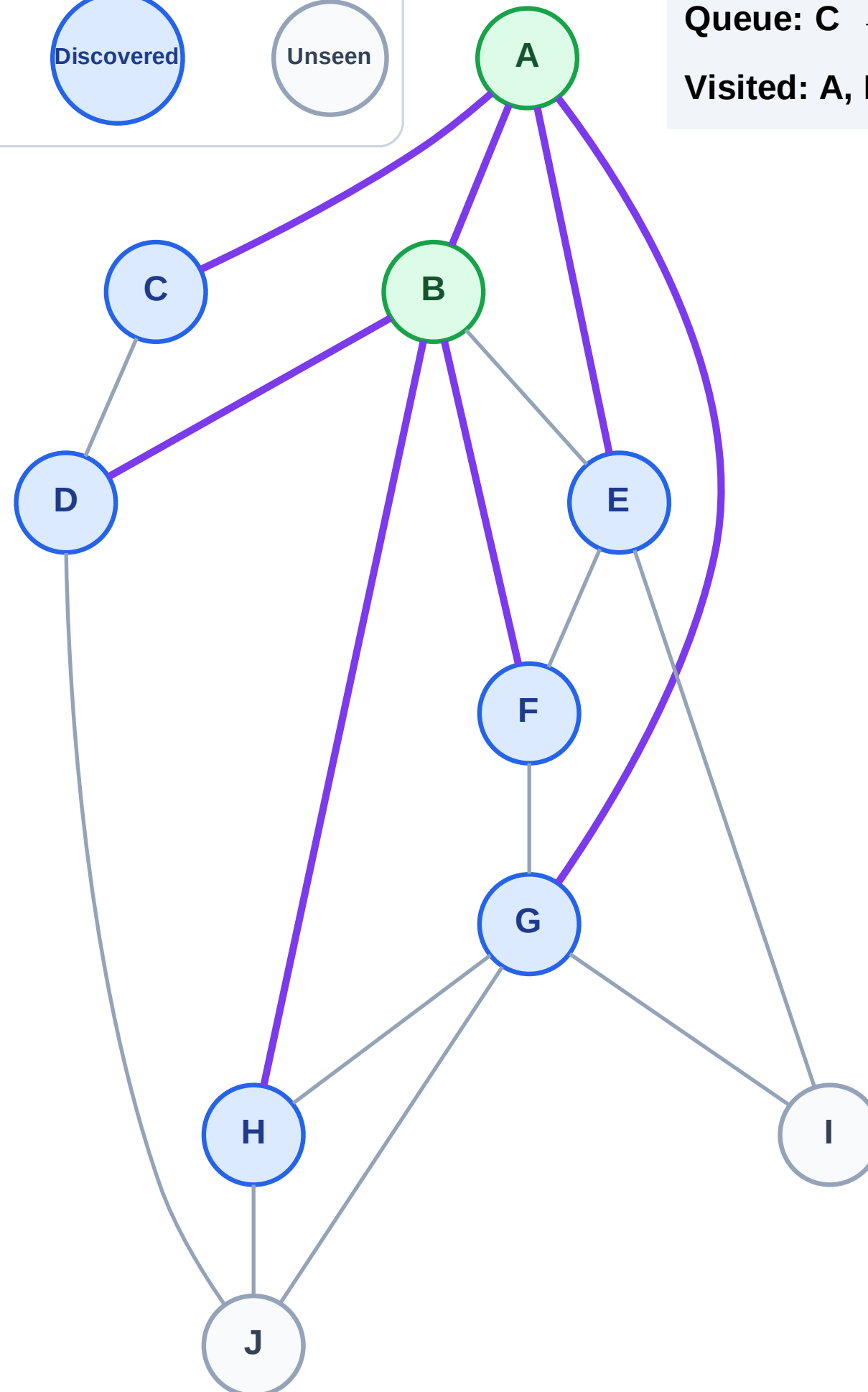
Step 5: Discover D, F, H from B

Legend



Queue: C → E → G → D → F → H

Visited: A, B



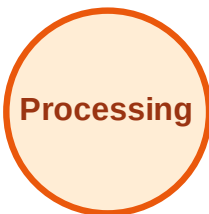
BFS Traversal

Step 6: Process node C

Legend



Complete



Processing



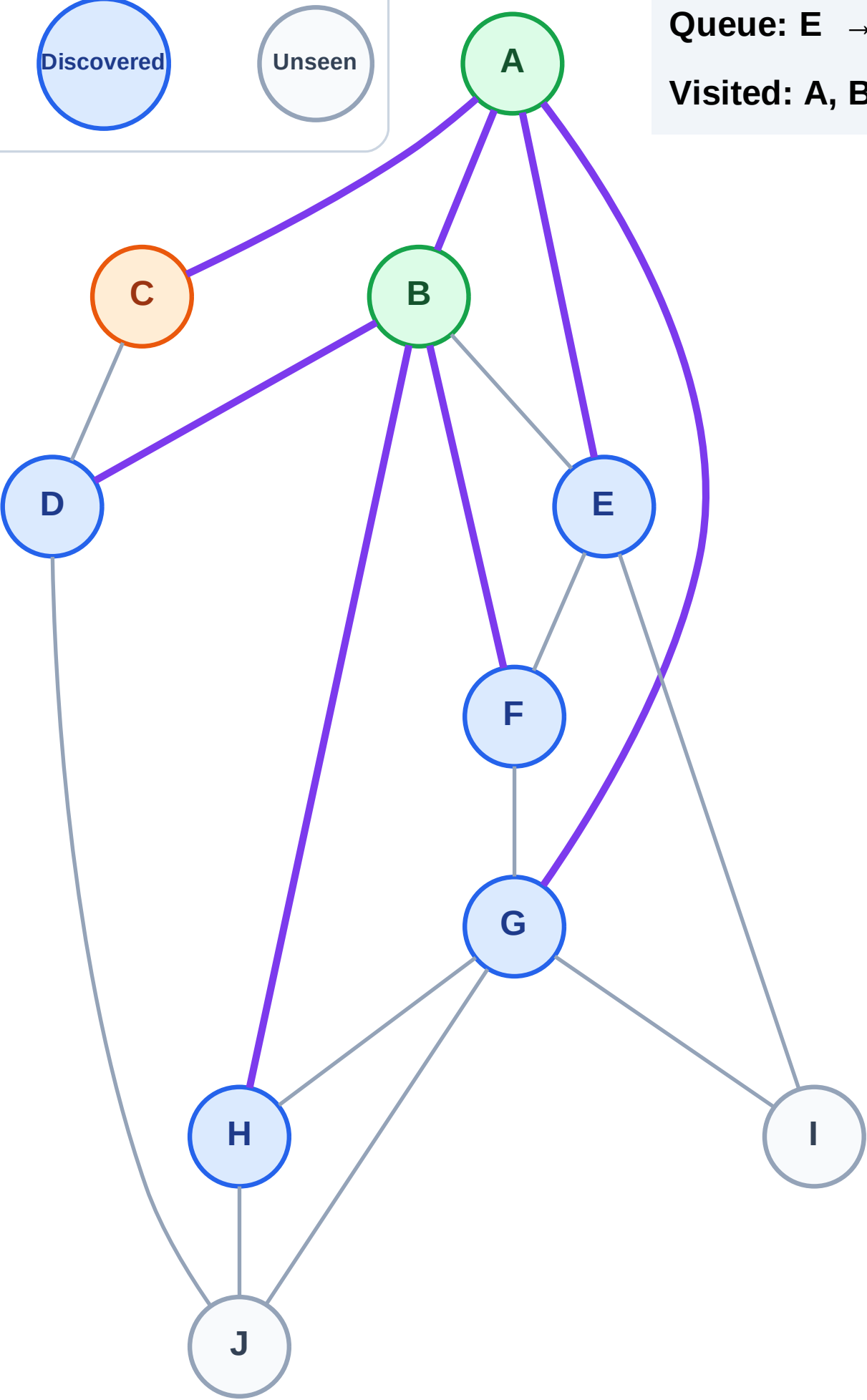
Discovered



Unseen

Queue: E → G → D → F → H

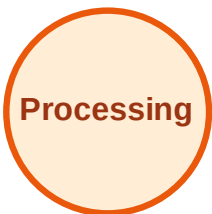
Visited: A, B



BFS Traversal

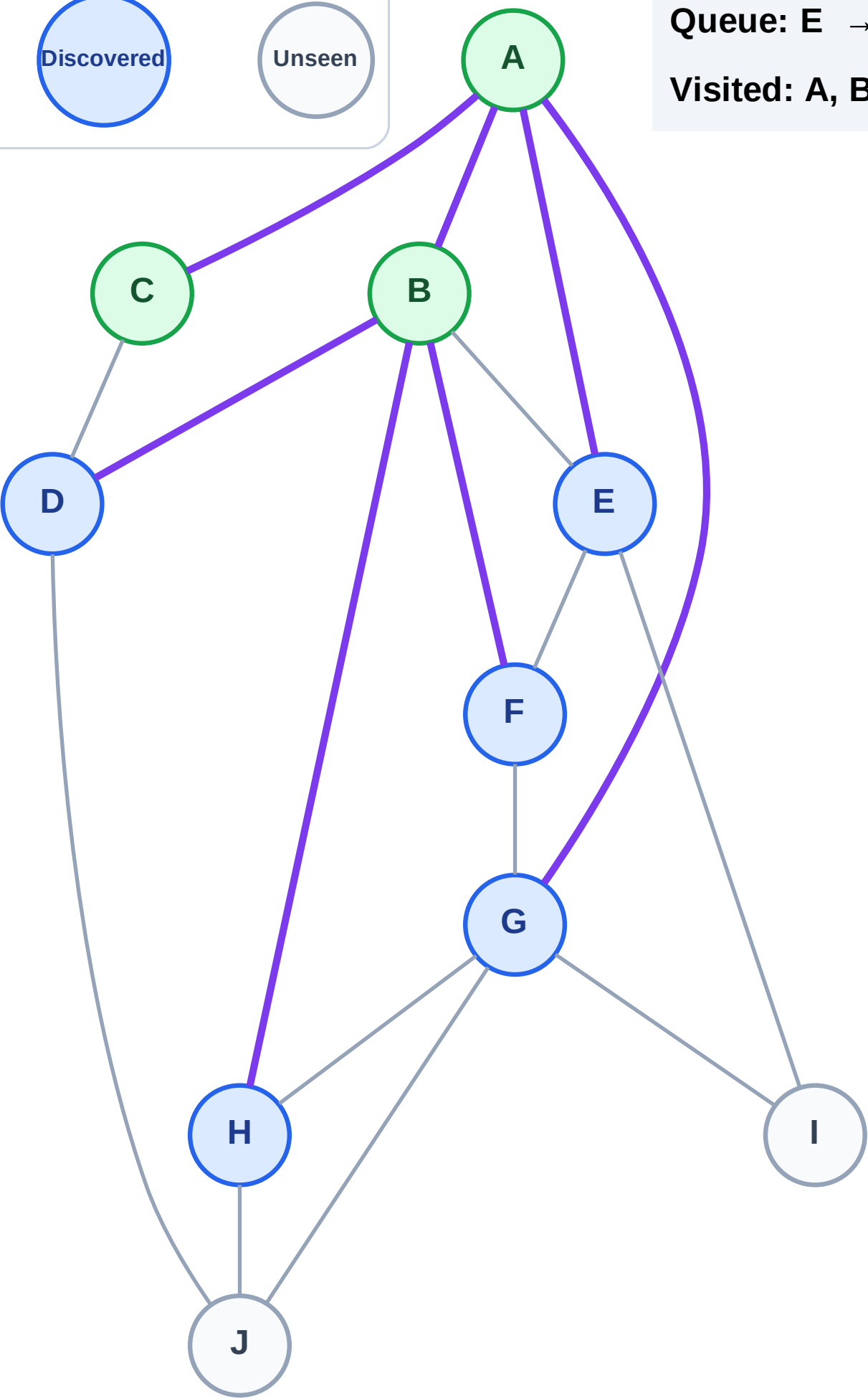
Step 7: No new neighbors from C

Legend



Queue: E → G → D → F → H

Visited: A, B, C



BFS Traversal

Step 8: Process node E

Legend

Complete

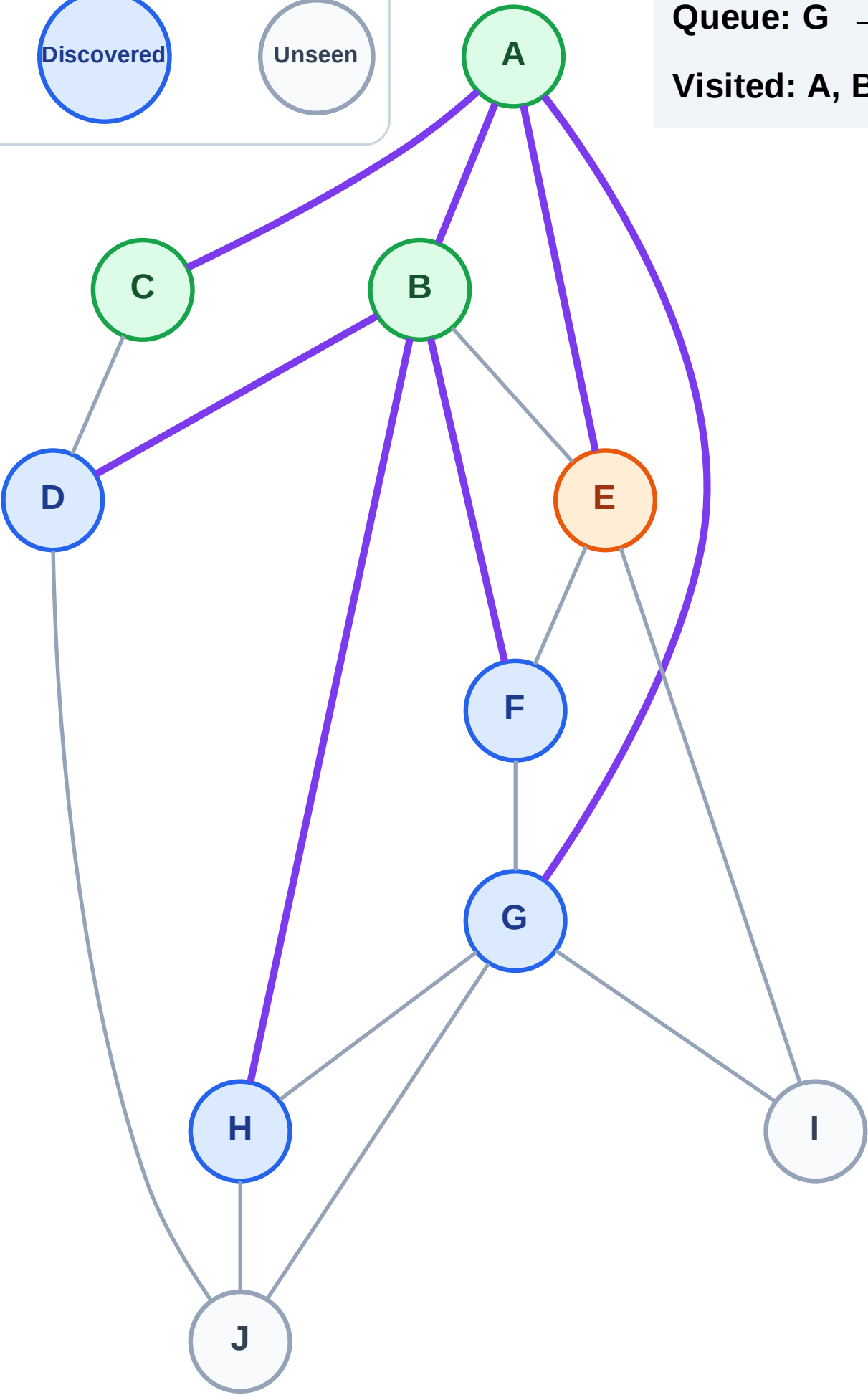
Processing

Discovered

Unseen

Queue: G → D → F → H

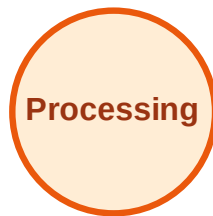
Visited: A, B, C



BFS Traversal

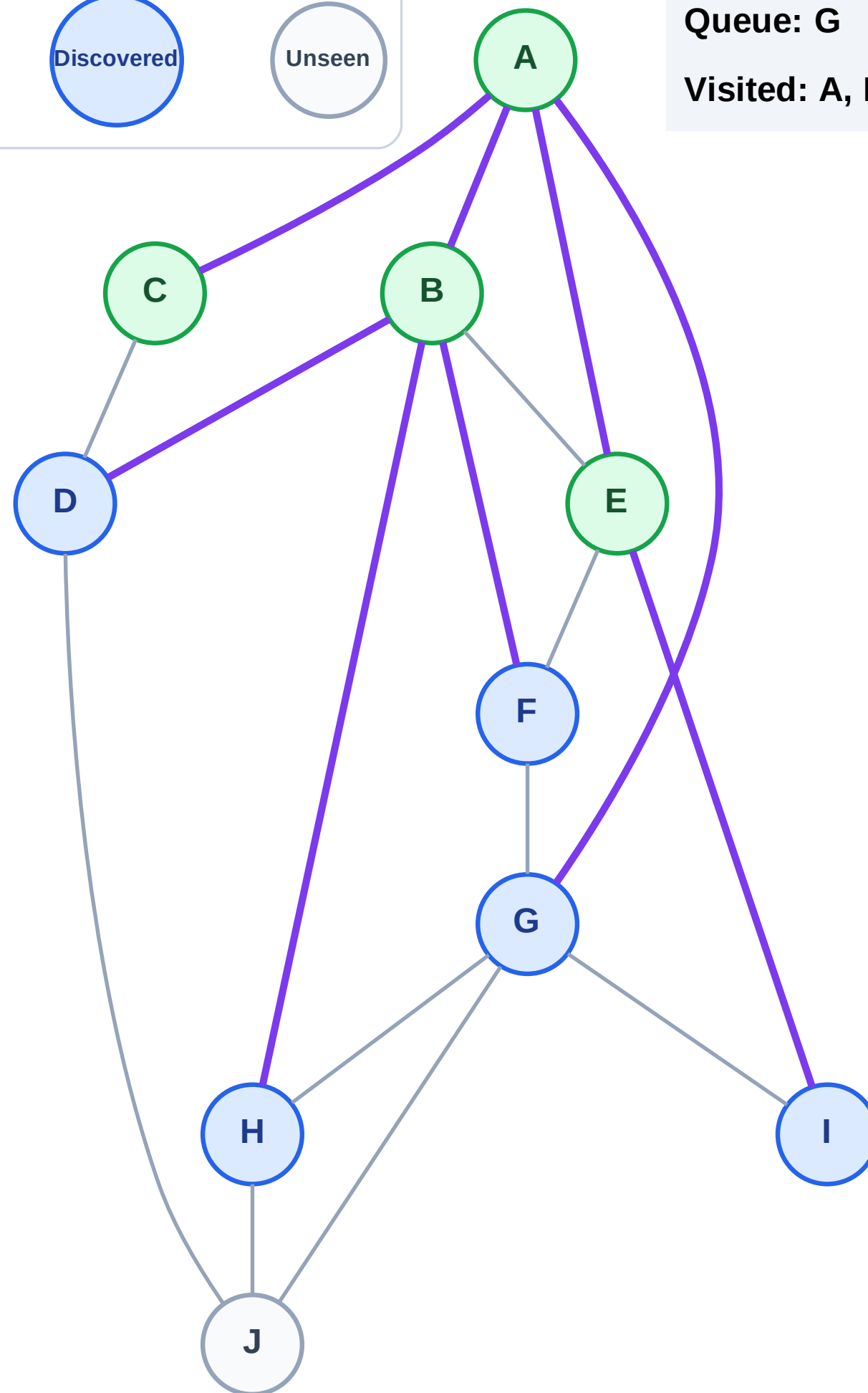
Step 9: Discover I from E

Legend



Queue: G → D → F → H → I

Visited: A, B, C, E



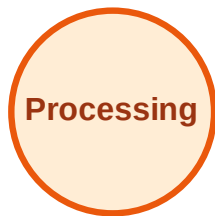
BFS Traversal

Step 10: Process node G

Legend



Complete



Processing

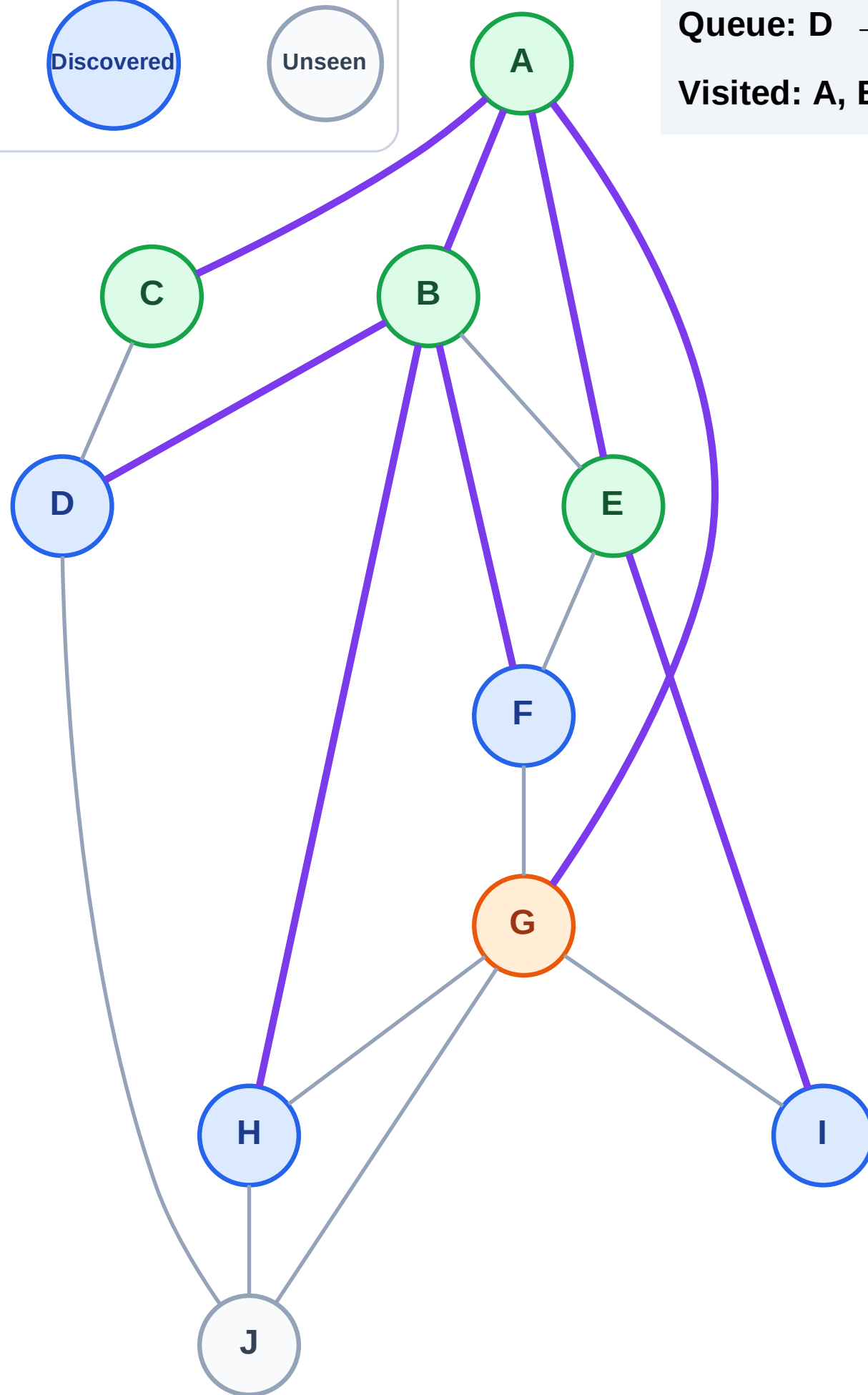


Discovered



Unseen

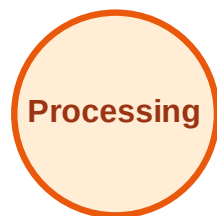
Queue: D → F → H → I
Visited: A, B, C, E



BFS Traversal

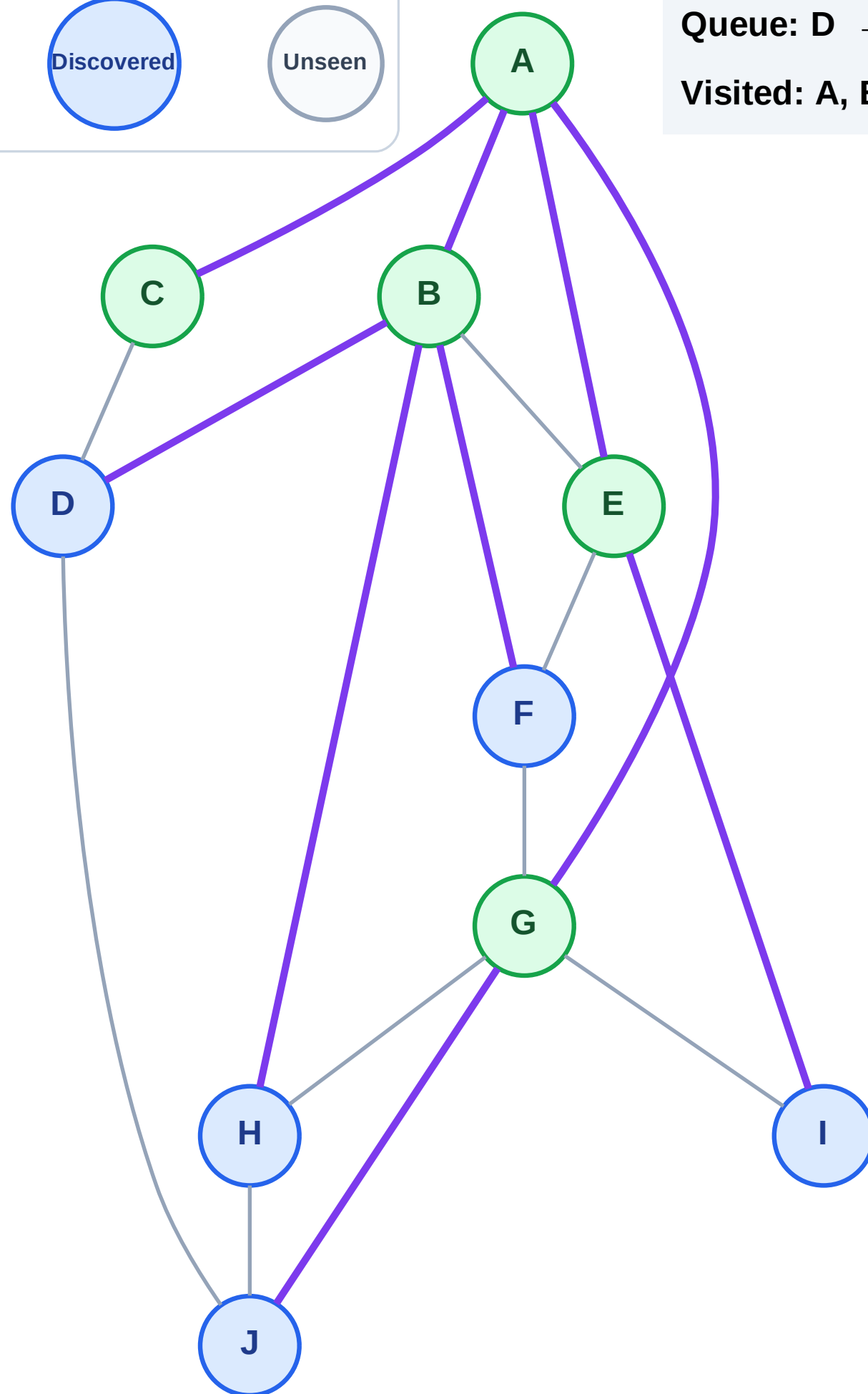
Step 11: Discover J from G

Legend



Queue: D → F → H → I → J

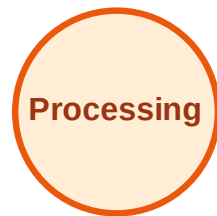
Visited: A, B, C, E, G



BFS Traversal

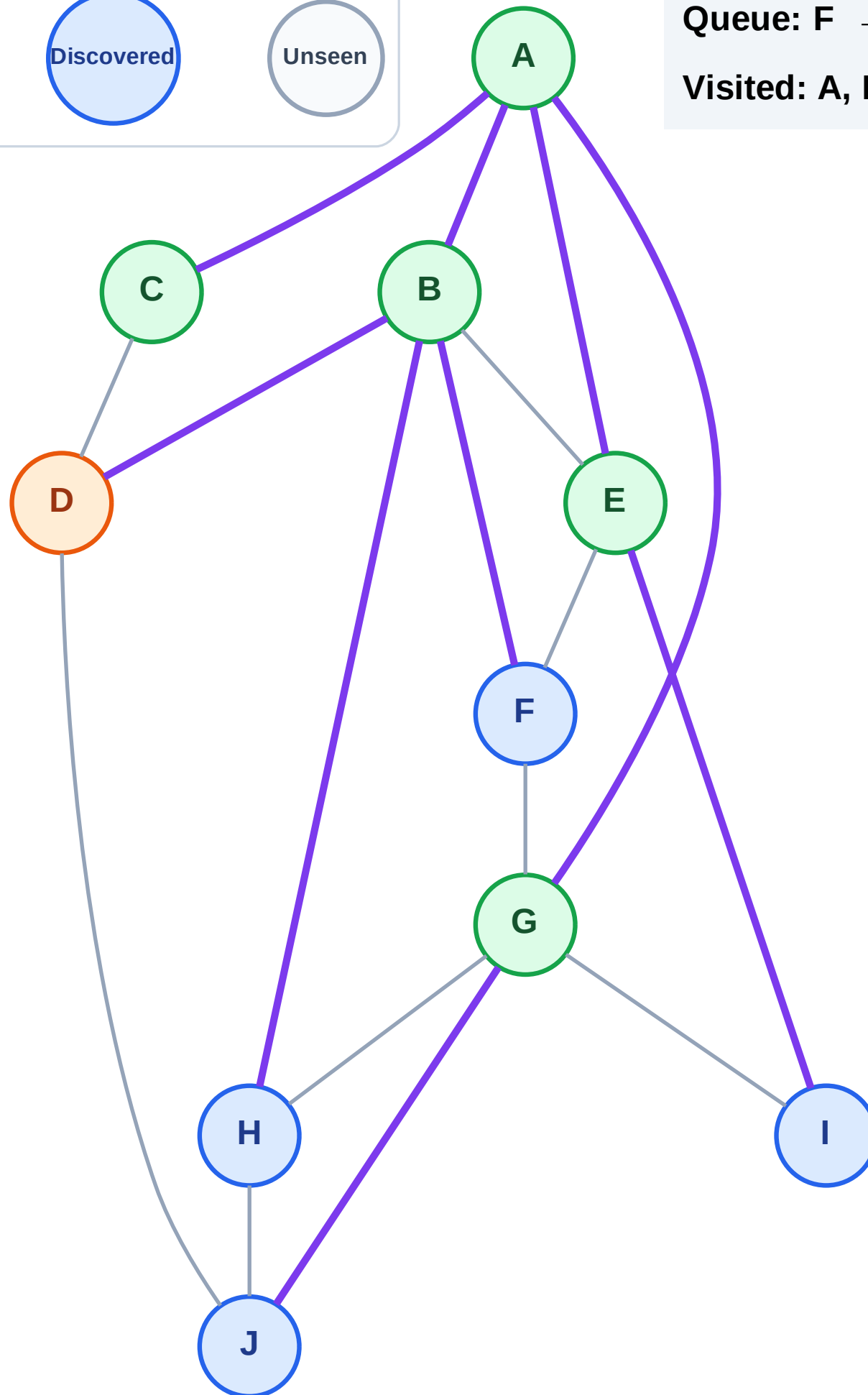
Step 12: Process node D

Legend



Queue: F → H → I → J

Visited: A, B, C, E, G



BFS Traversal

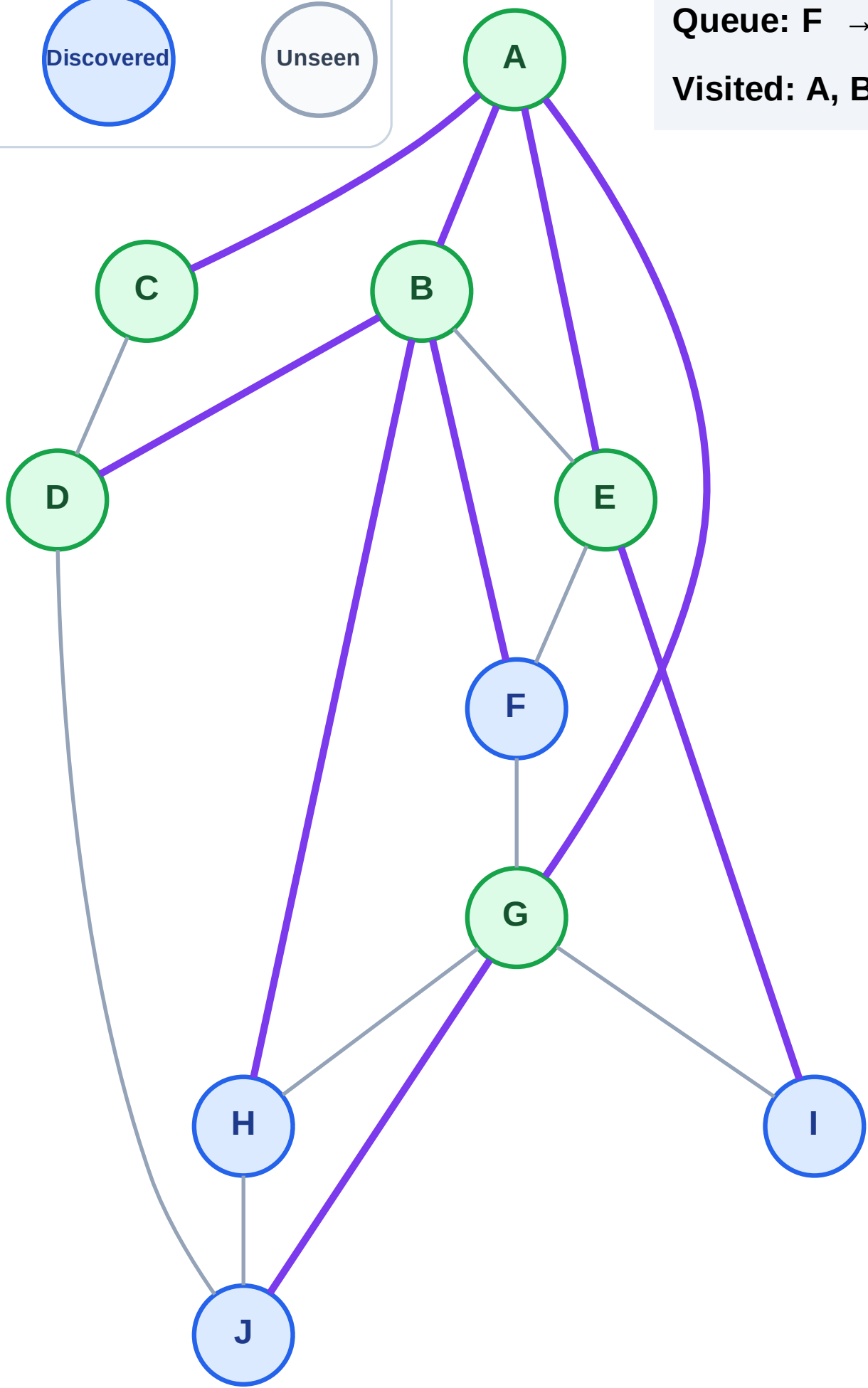
Step 13: No new neighbors from D

Legend



Queue: F → H → I → J

Visited: A, B, C, D, E, G



BFS Traversal

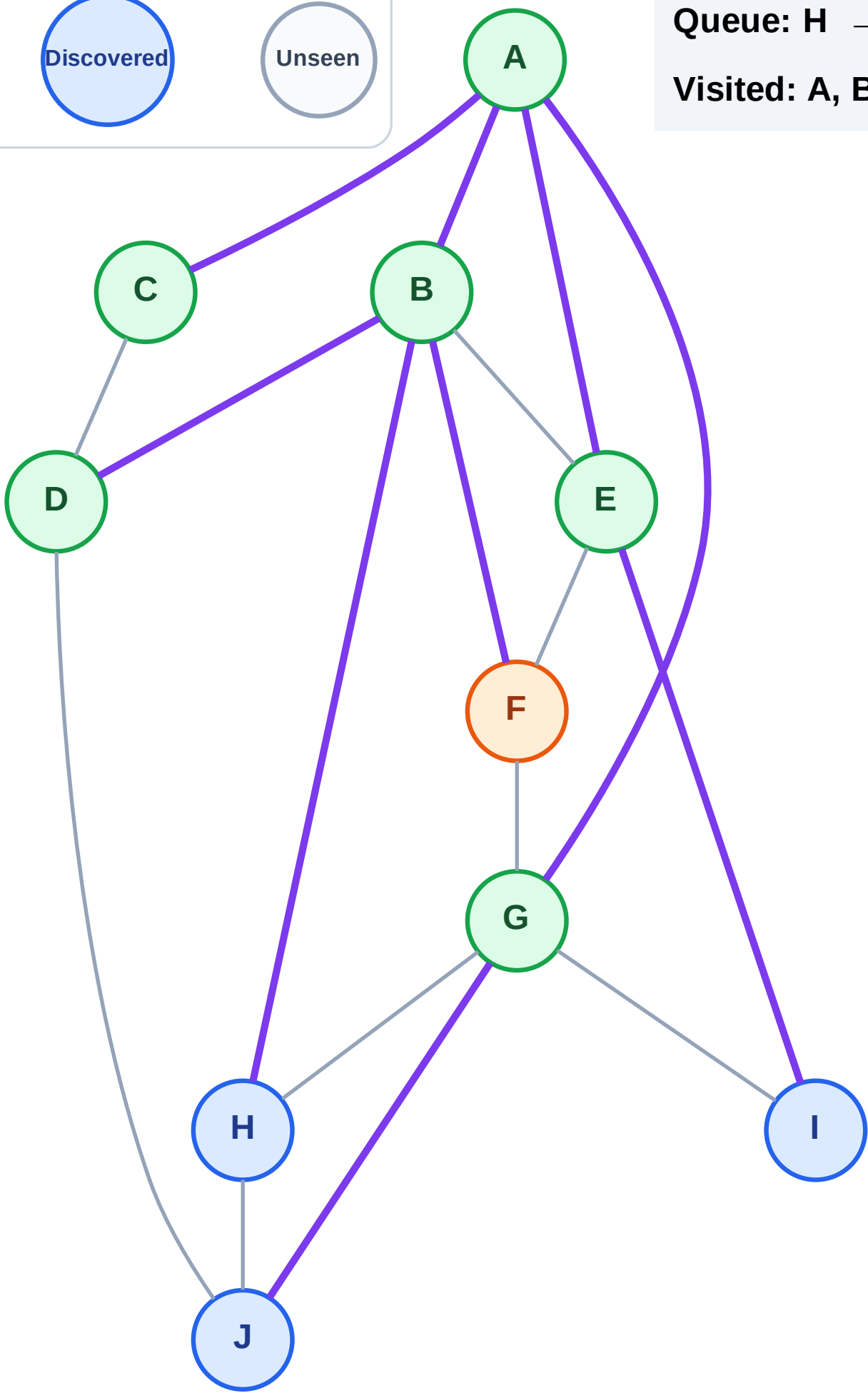
Step 14: Process node F

Legend



Queue: H → I → J

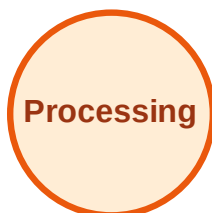
Visited: A, B, C, D, E, G



BFS Traversal

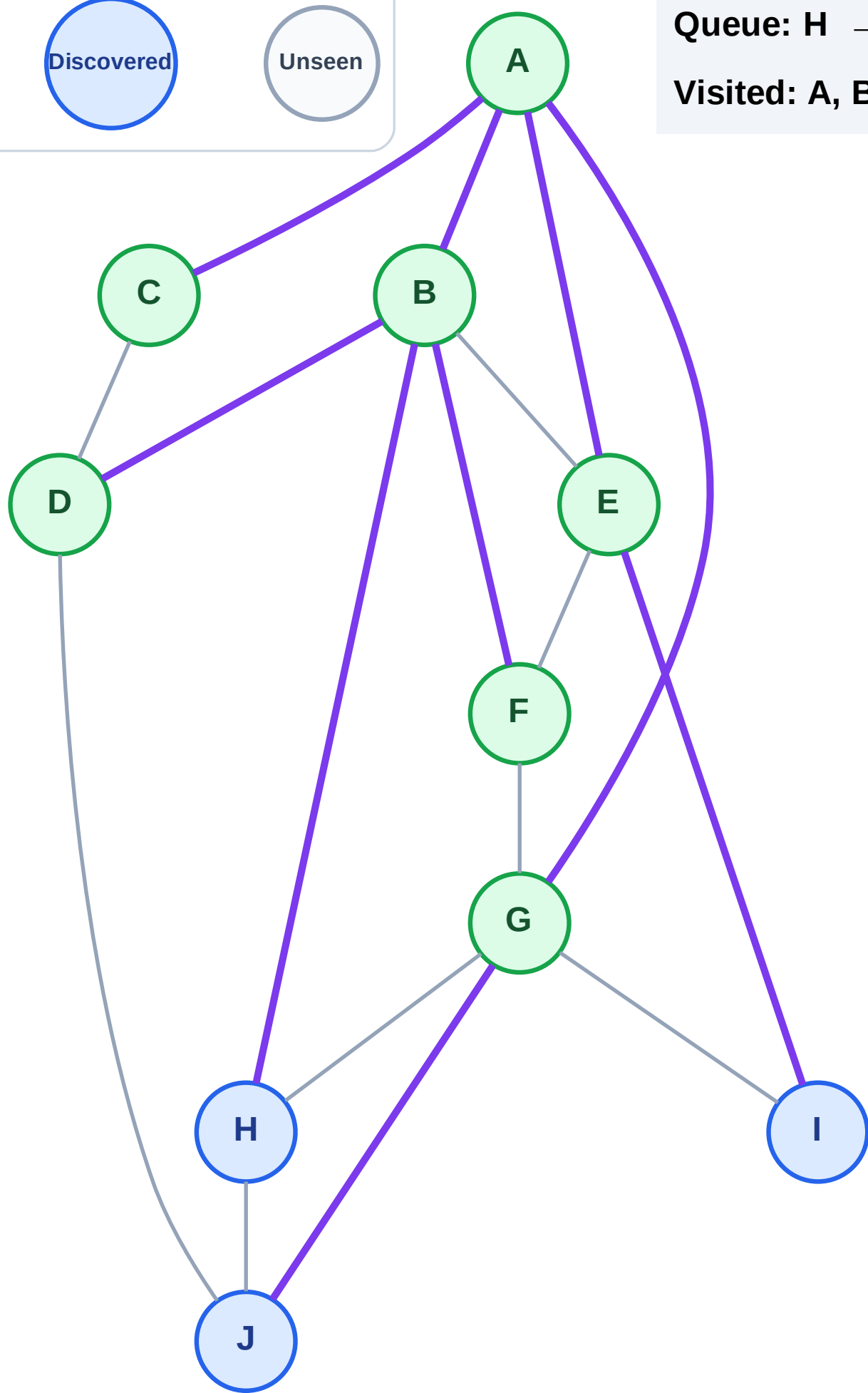
Step 15: No new neighbors from F

Legend



Queue: H → I → J

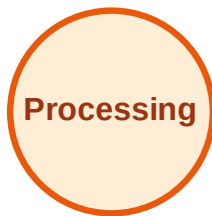
Visited: A, B, C, D, E, F, G



BFS Traversal

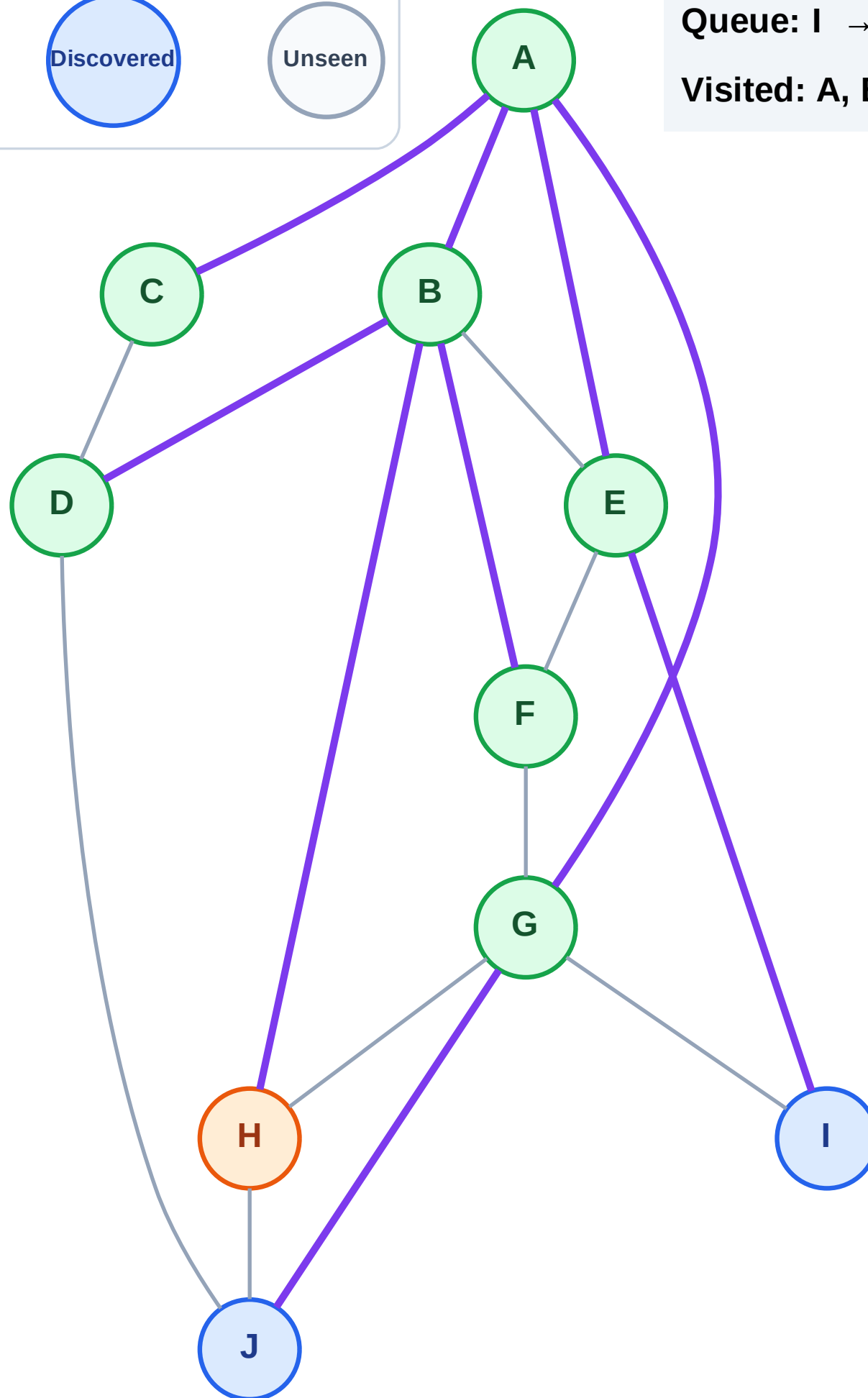
Step 16: Process node H

Legend



Queue: I → J

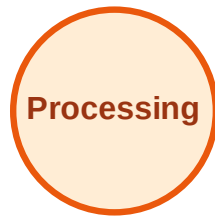
Visited: A, B, C, D, E, F, G



BFS Traversal

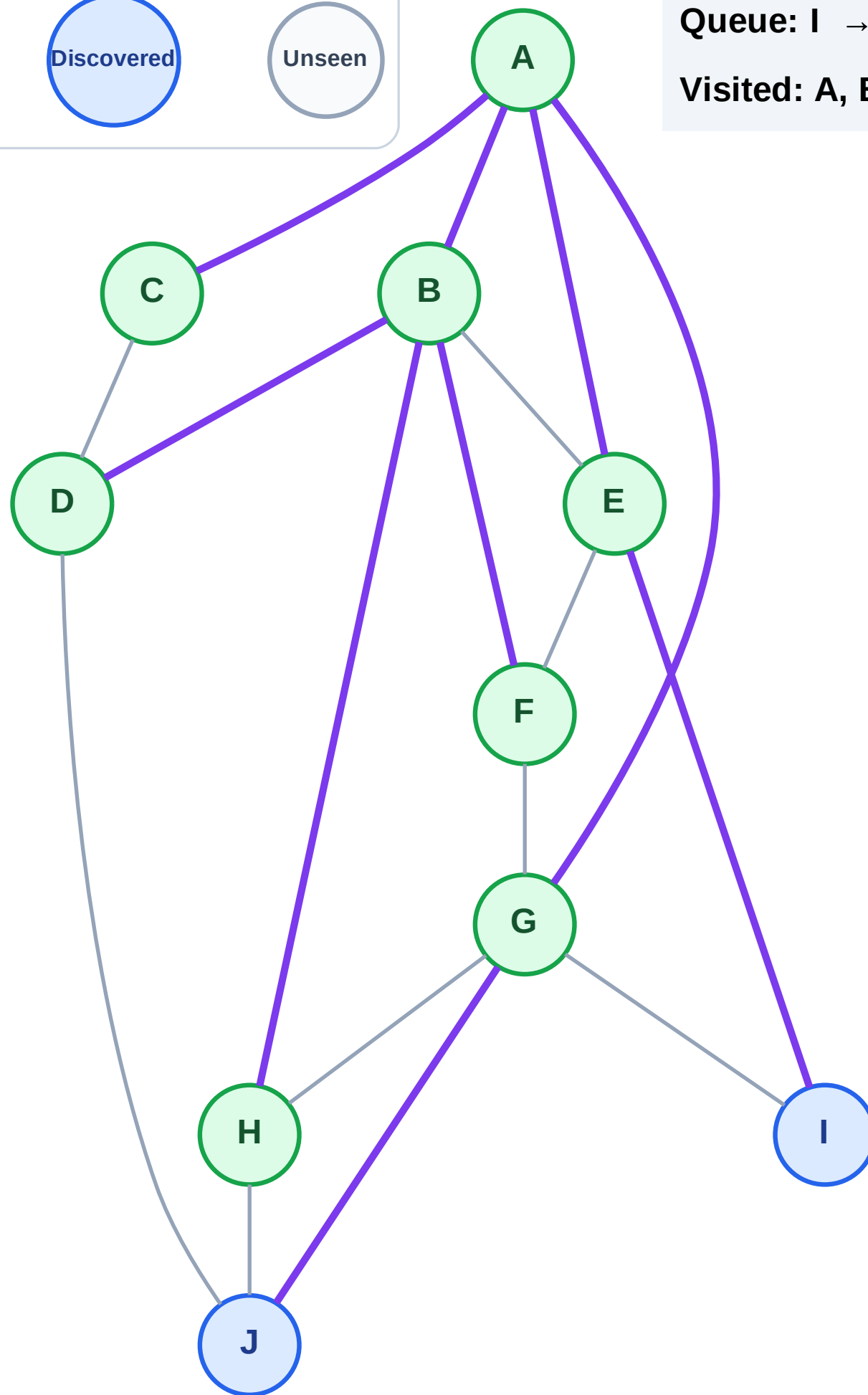
Step 17: No new neighbors from H

Legend



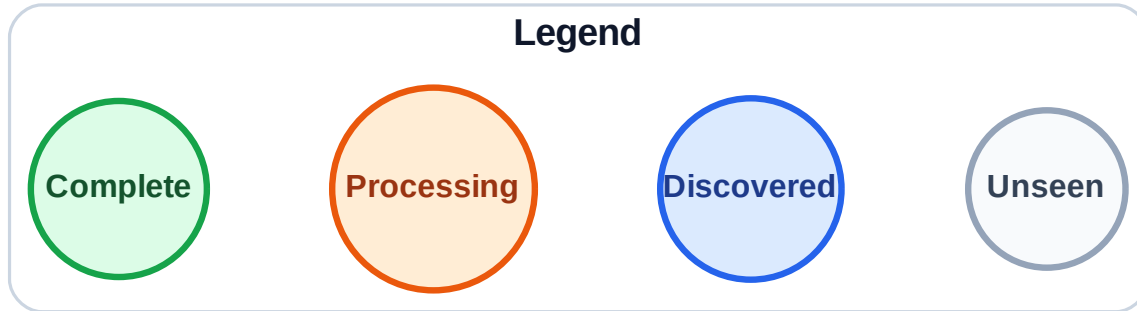
Queue: I → J

Visited: A, B, C, D, E, F, G, H



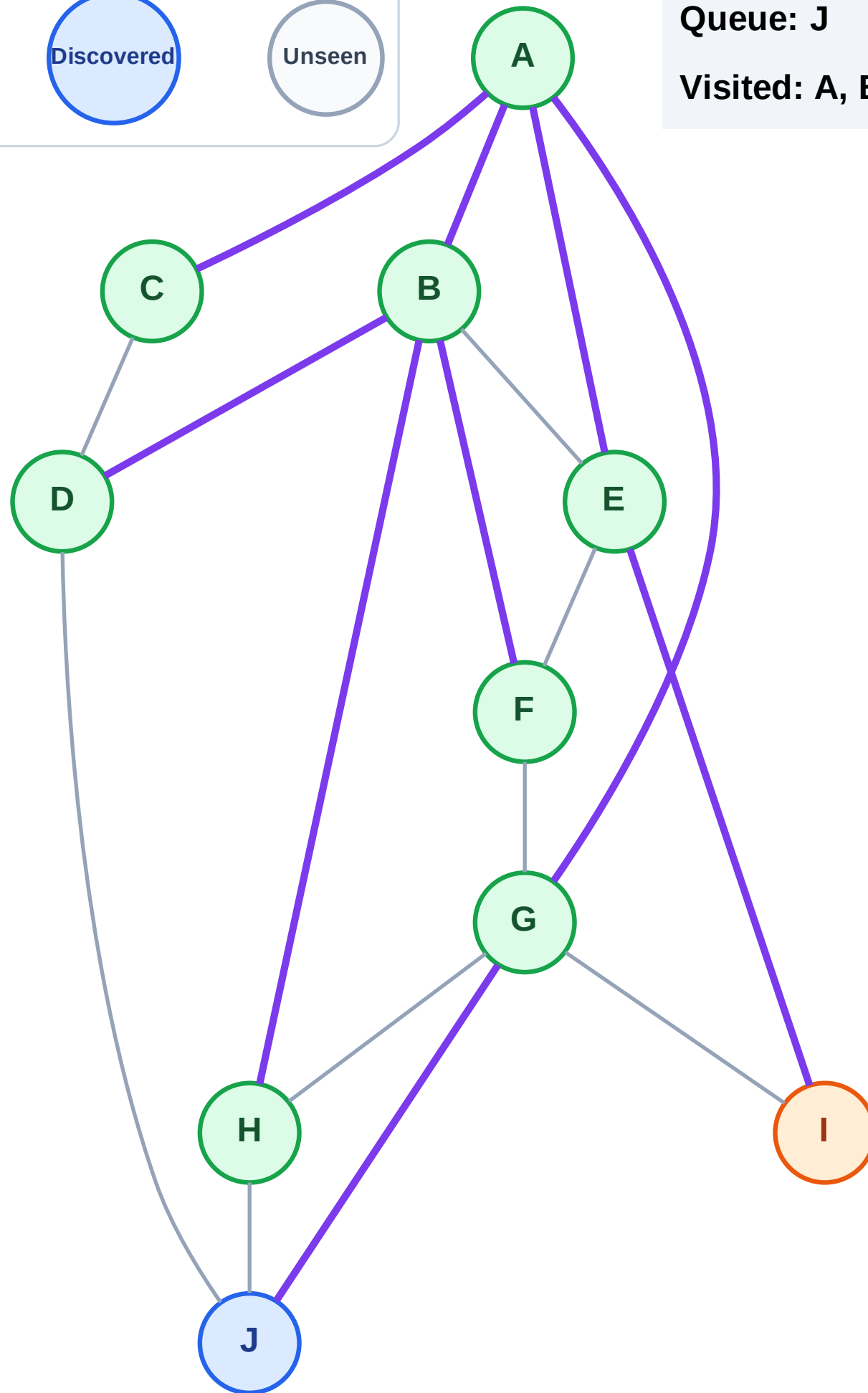
Step 18: Process node I

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Queue: J

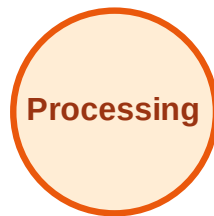
Visited: A, B, C, D, E, F, G, H



BFS Traversal

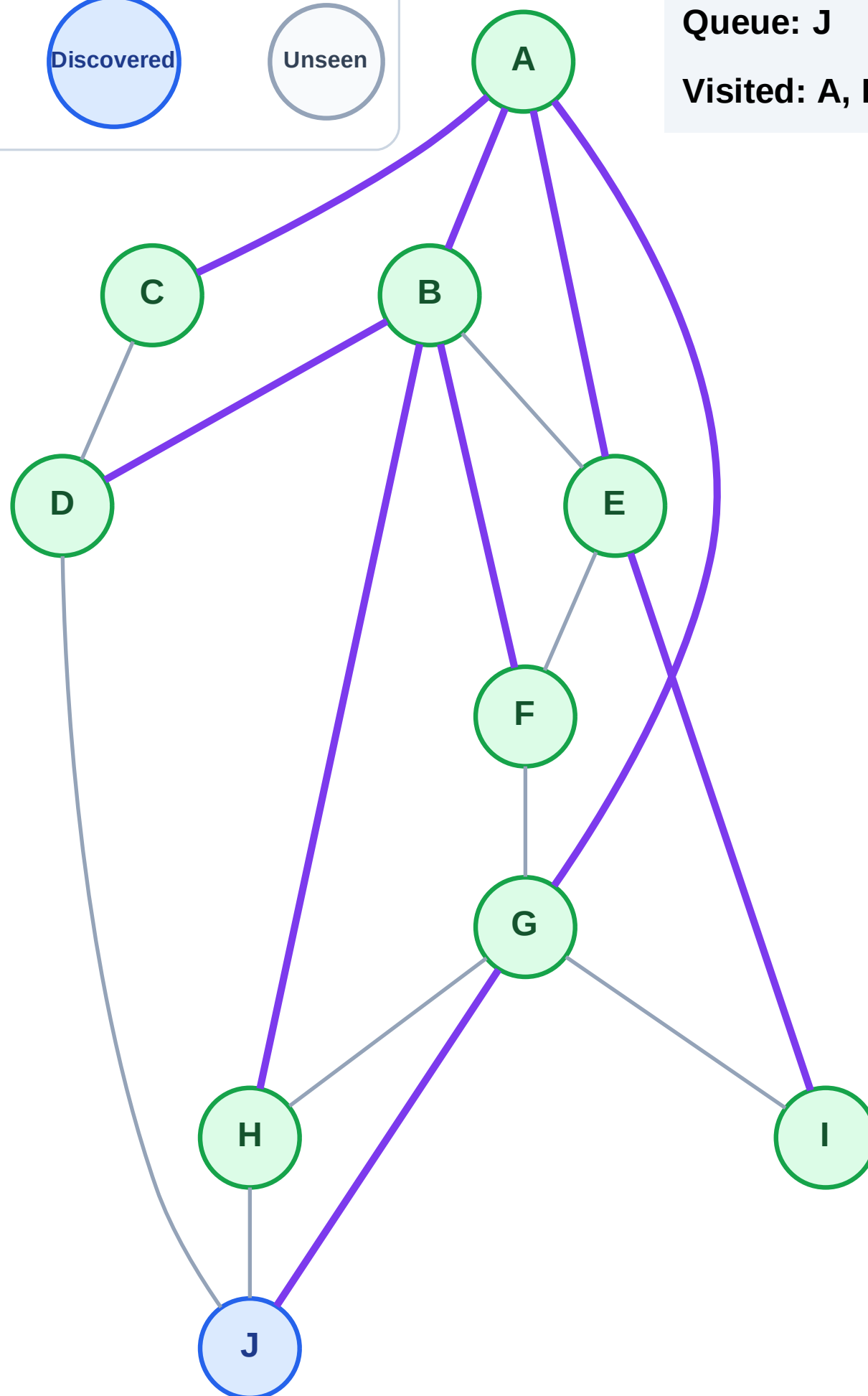
Step 19: No new neighbors from I

Legend



Queue: J

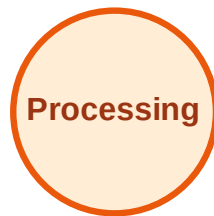
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

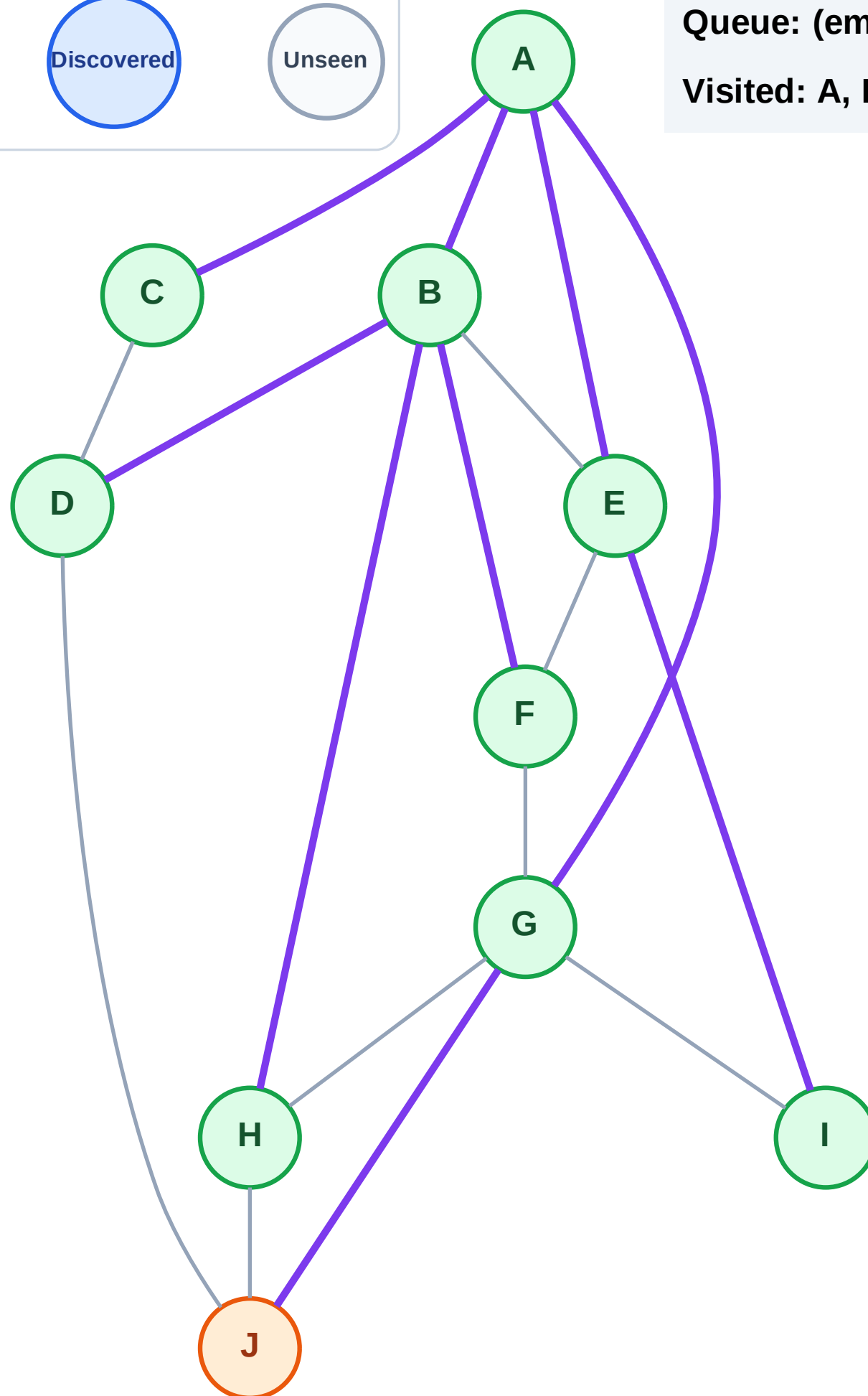
Step 20: Process node J

Legend



Queue: (empty)

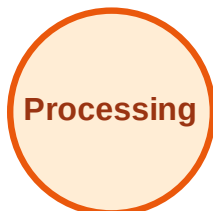
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

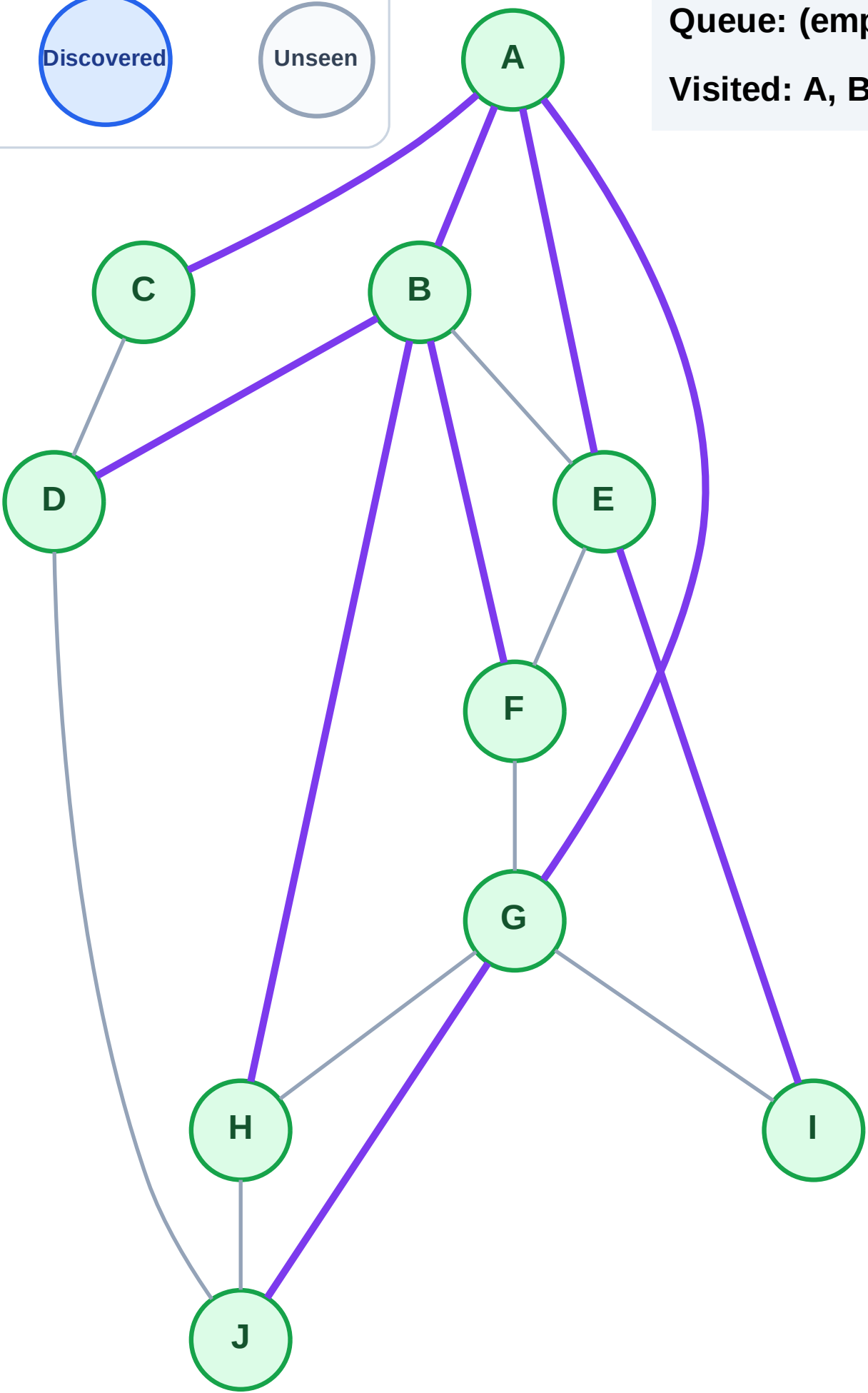
Step 21: No new neighbors from J

Legend



Queue: (empty)

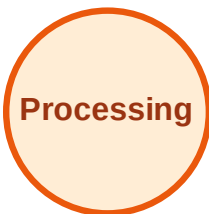
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

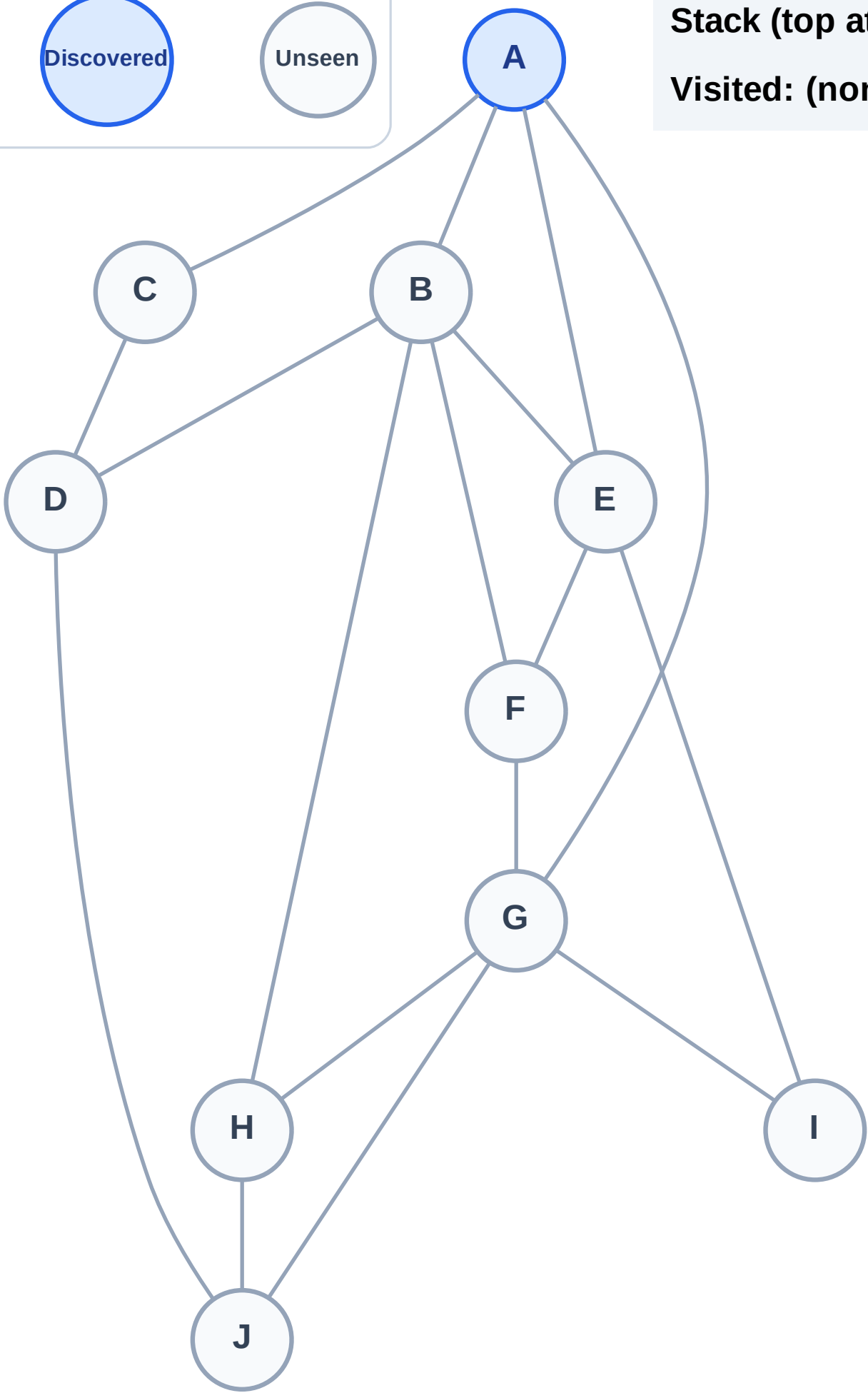
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

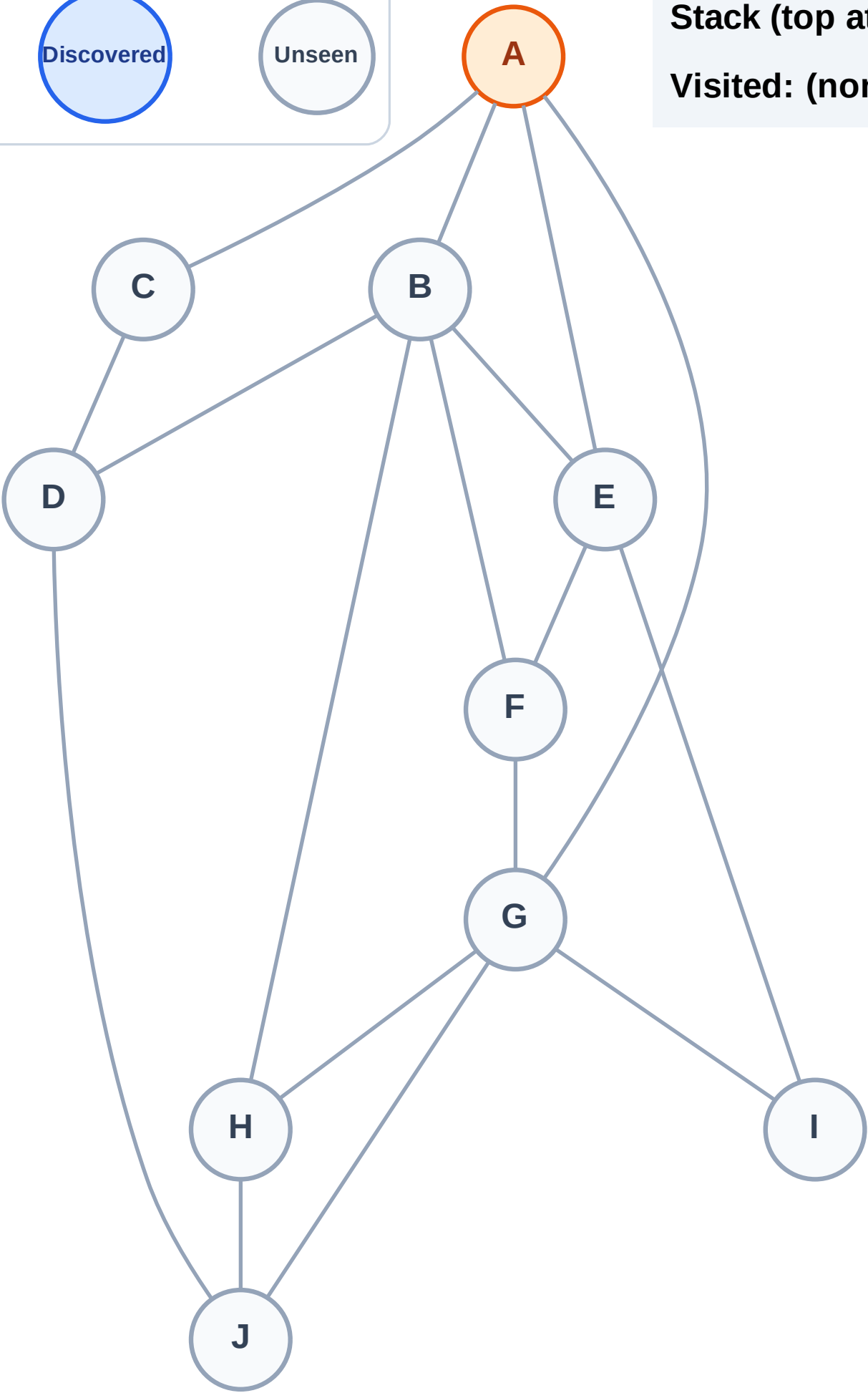
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover G, E, C, B from A

Legend

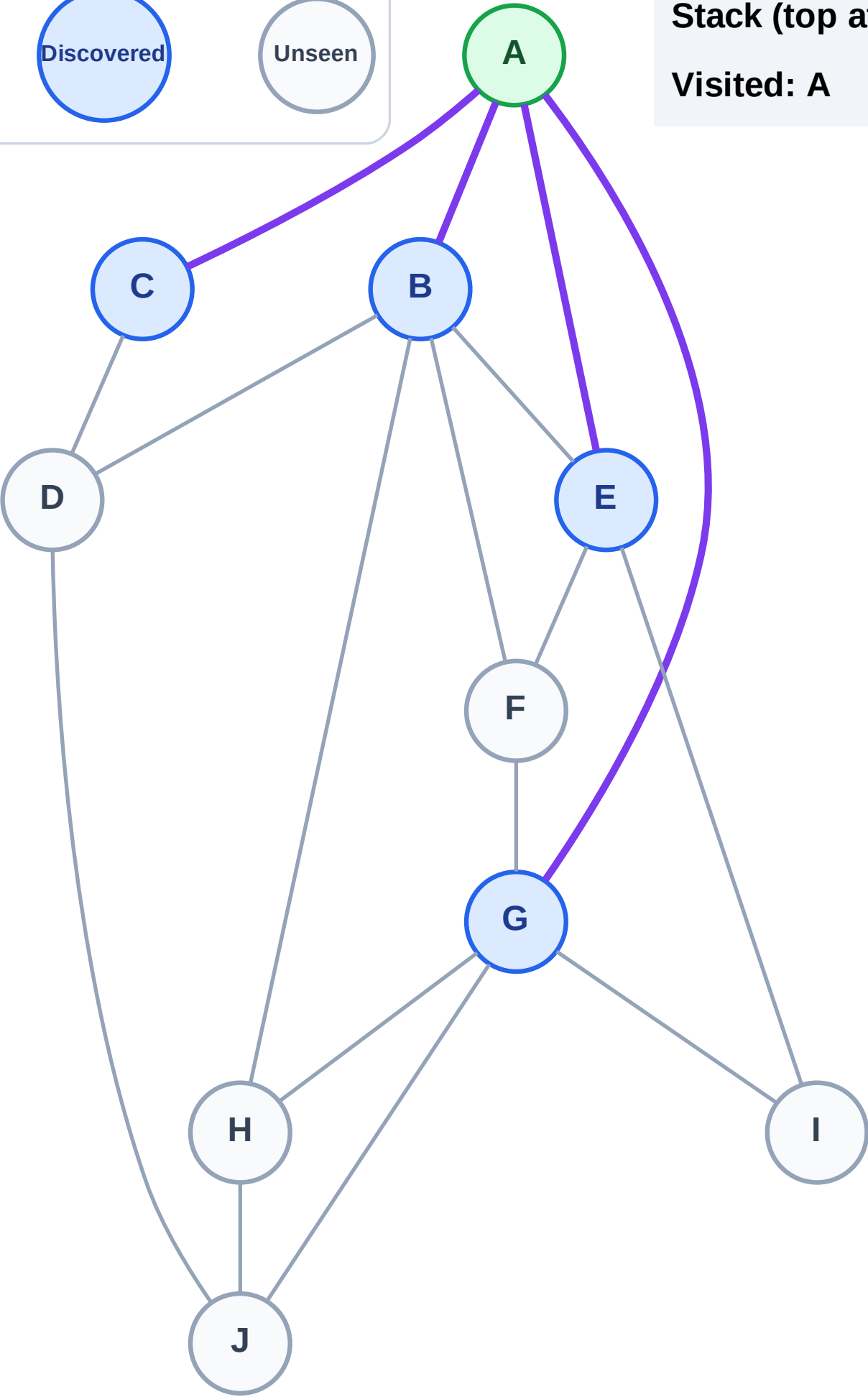
Complete

Processing

Discovered

Unseen

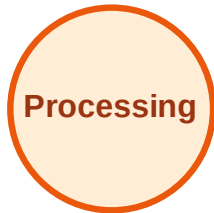
Stack (top at right): G | E | C | B
Visited: A



DFS Traversal

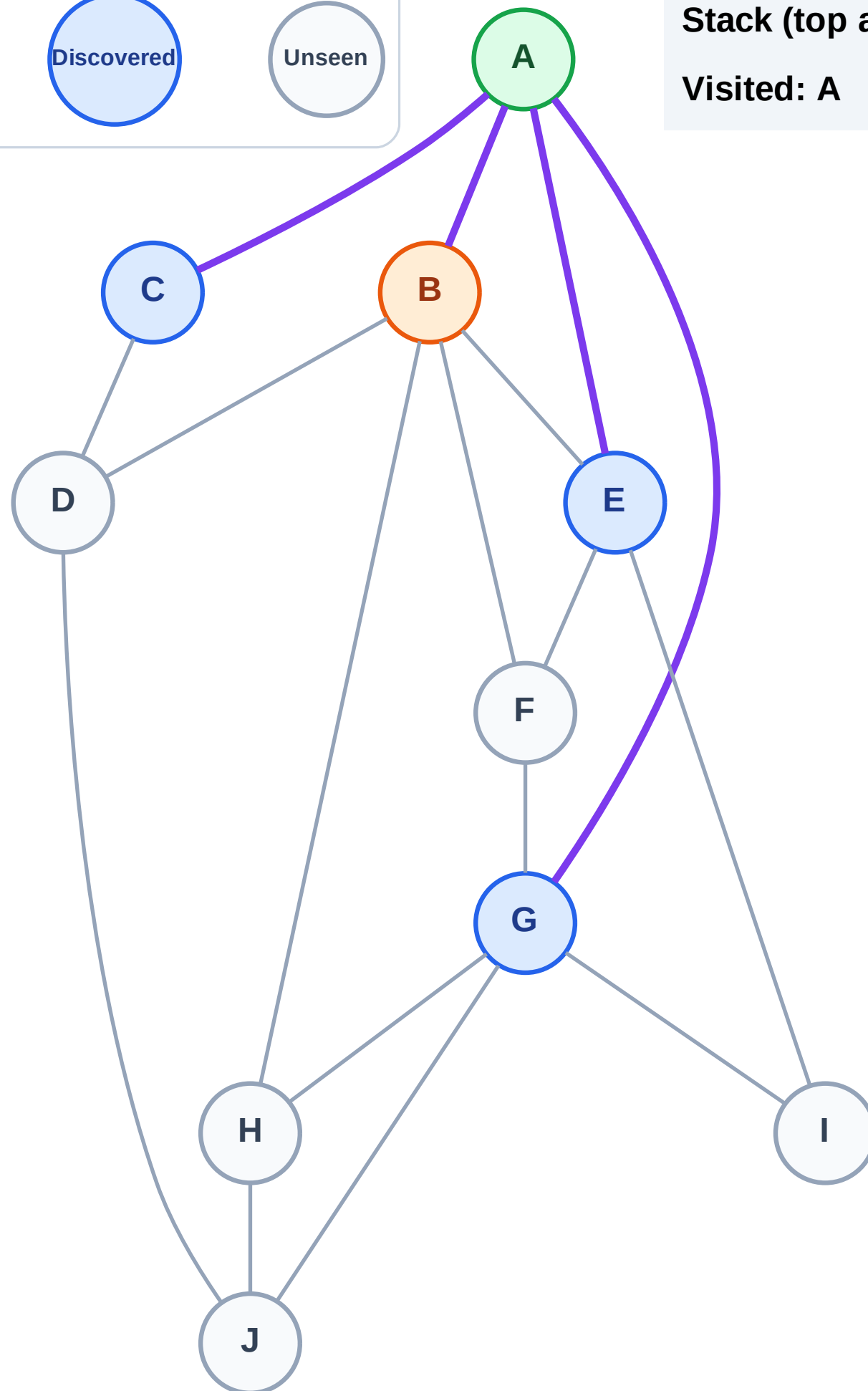
Step 4: Process node B

Legend



Stack (top at right): G | E | C

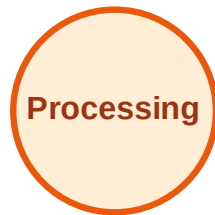
Visited: A



DFS Traversal

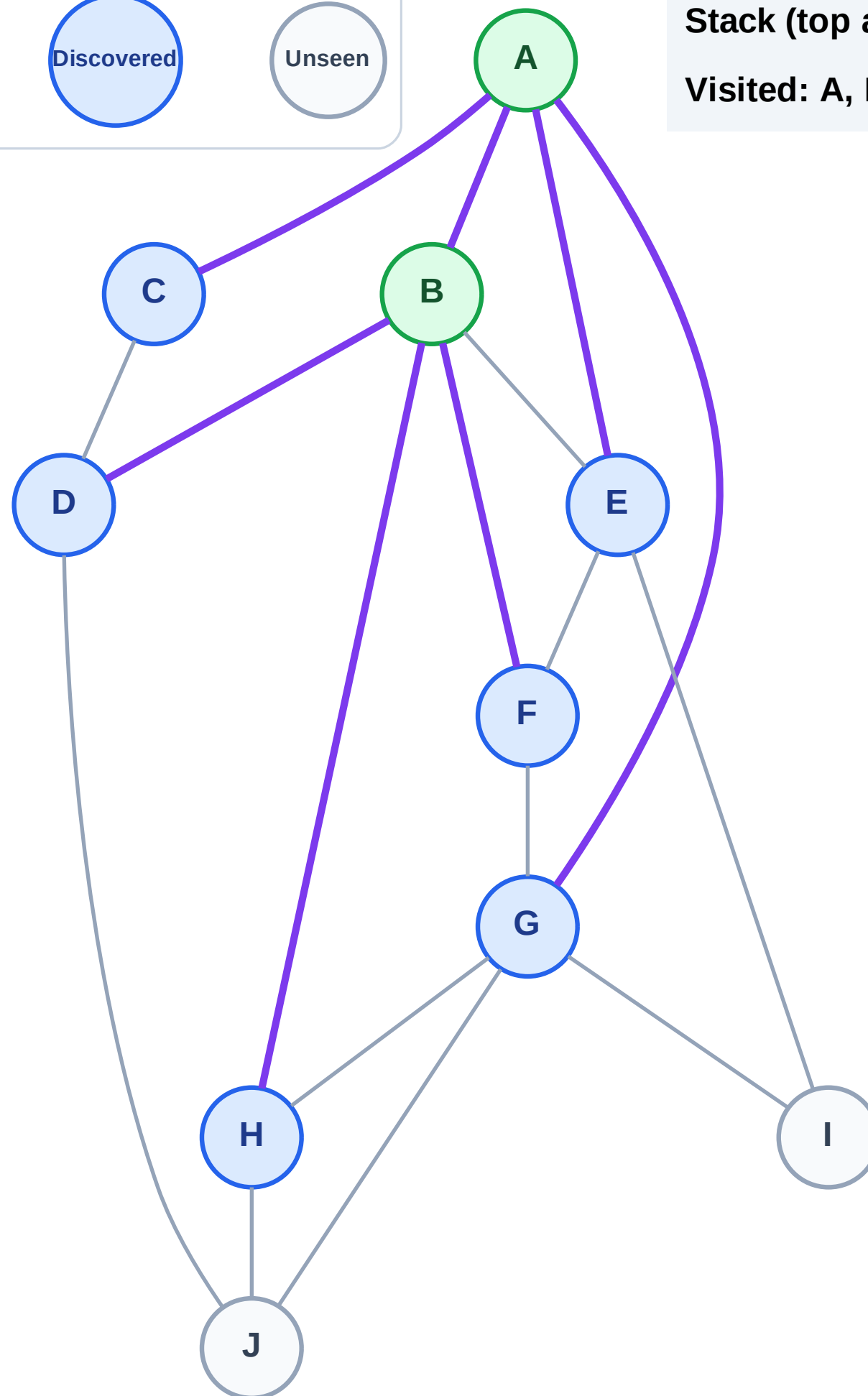
Step 5: Discover H, F, D from B

Legend



Stack (top at right): G | E | C | H | F | D

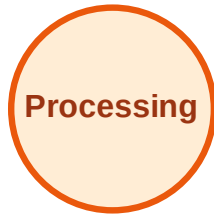
Visited: A, B



DFS Traversal

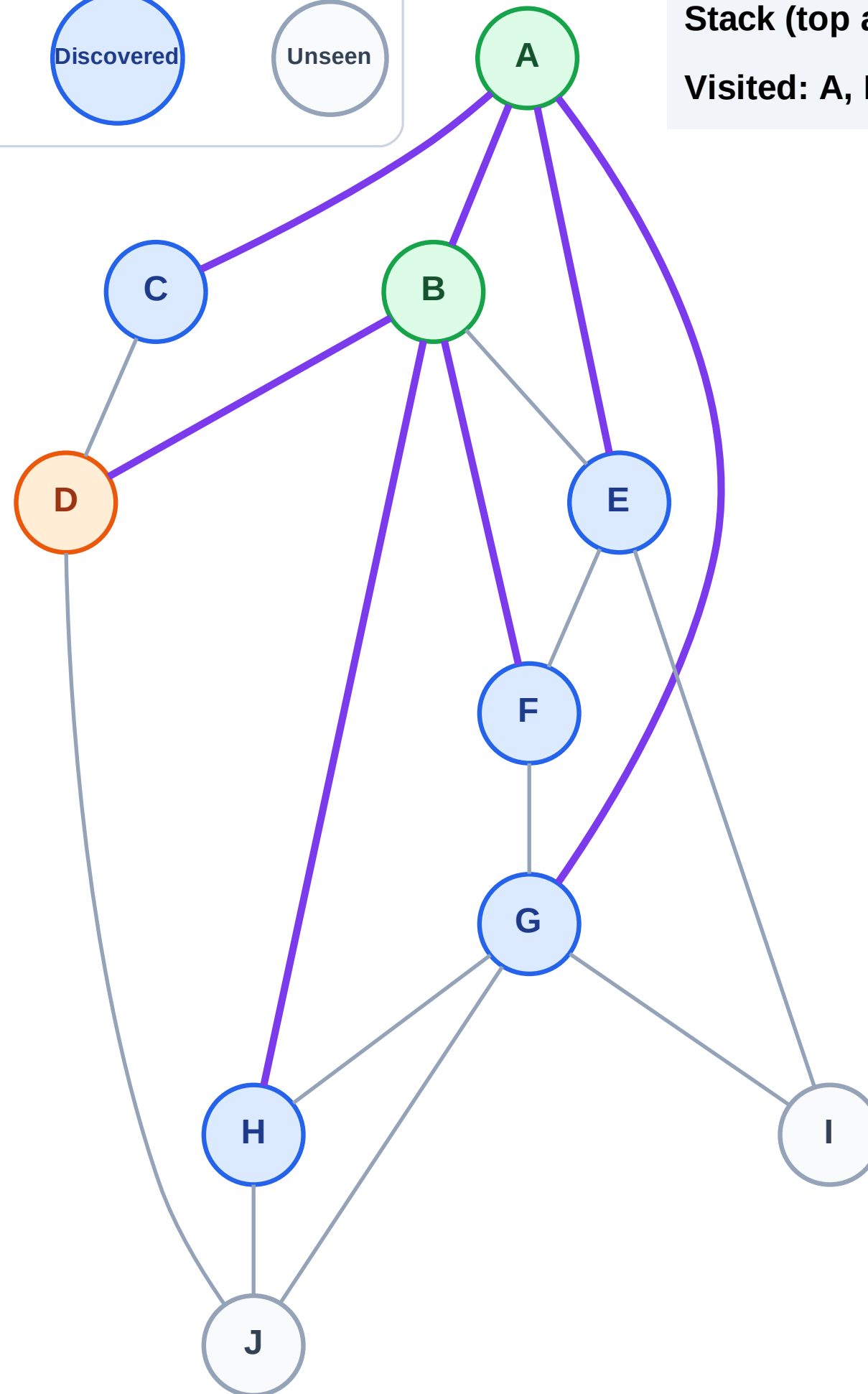
Step 6: Process node D

Legend



Stack (top at right): G | E | C | H | F

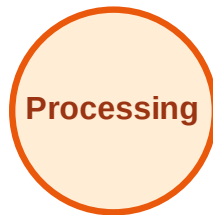
Visited: A, B



DFS Traversal

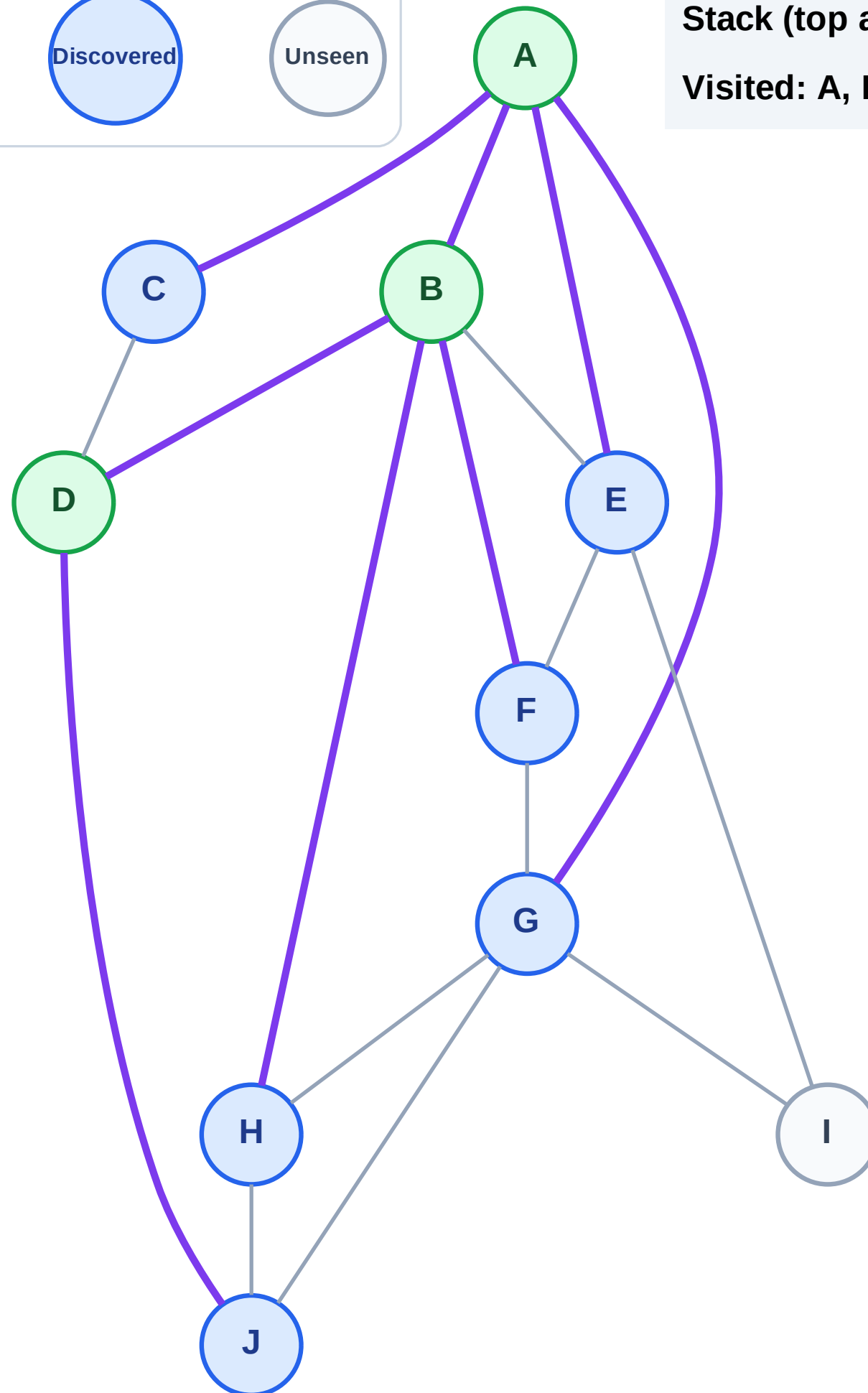
Step 7: Discover J from D

Legend



Stack (top at right): G | E | C | H | F | J

Visited: A, B, D



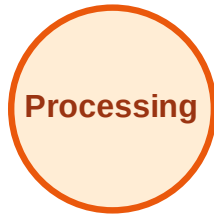
DFS Traversal

Step 8: Process node J

Legend



Complete



Processing



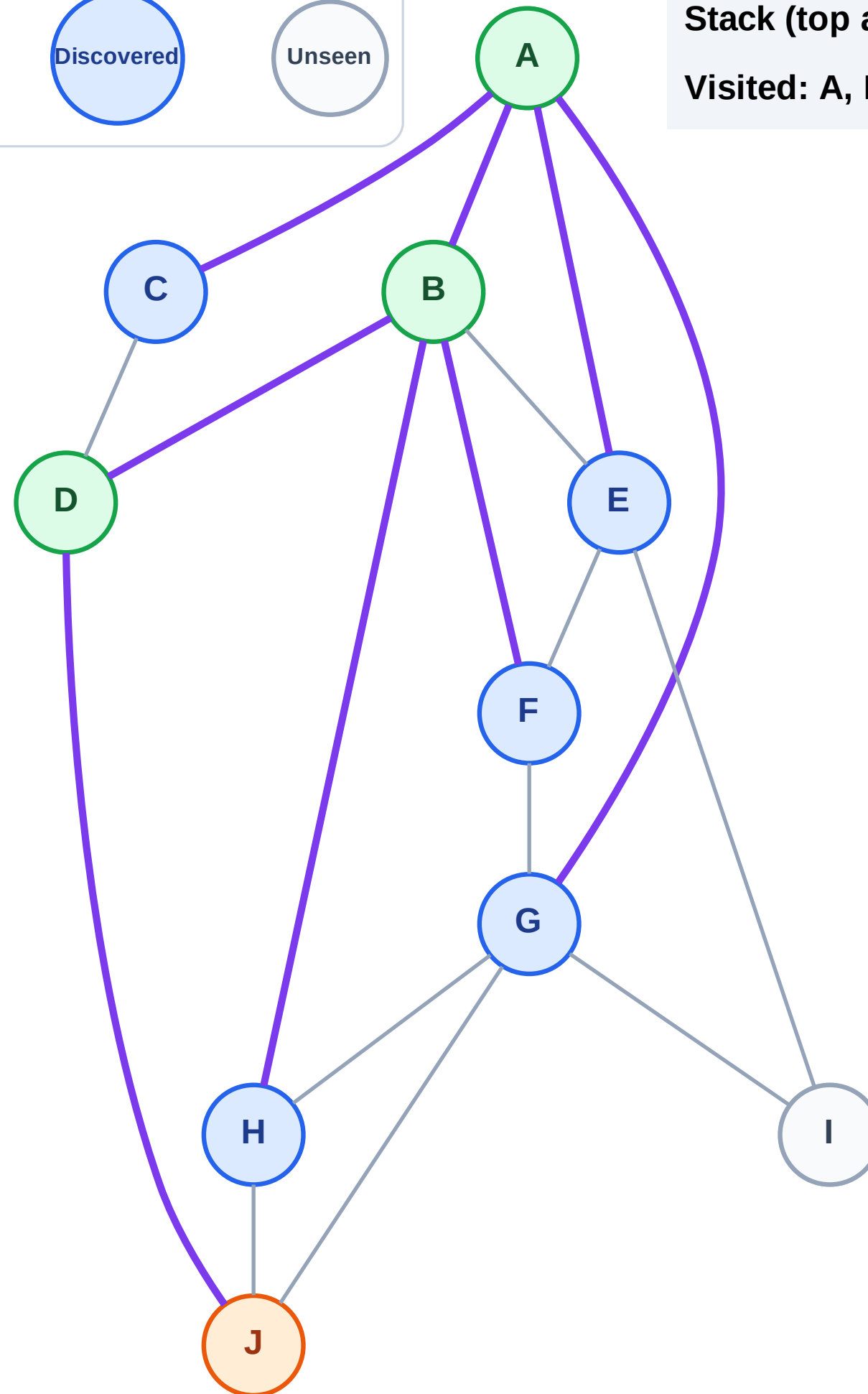
Discovered



Unseen

Stack (top at right): G | E | C | H | F

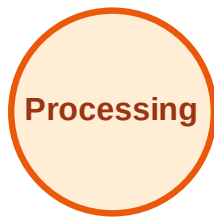
Visited: A, B, D



DFS Traversal

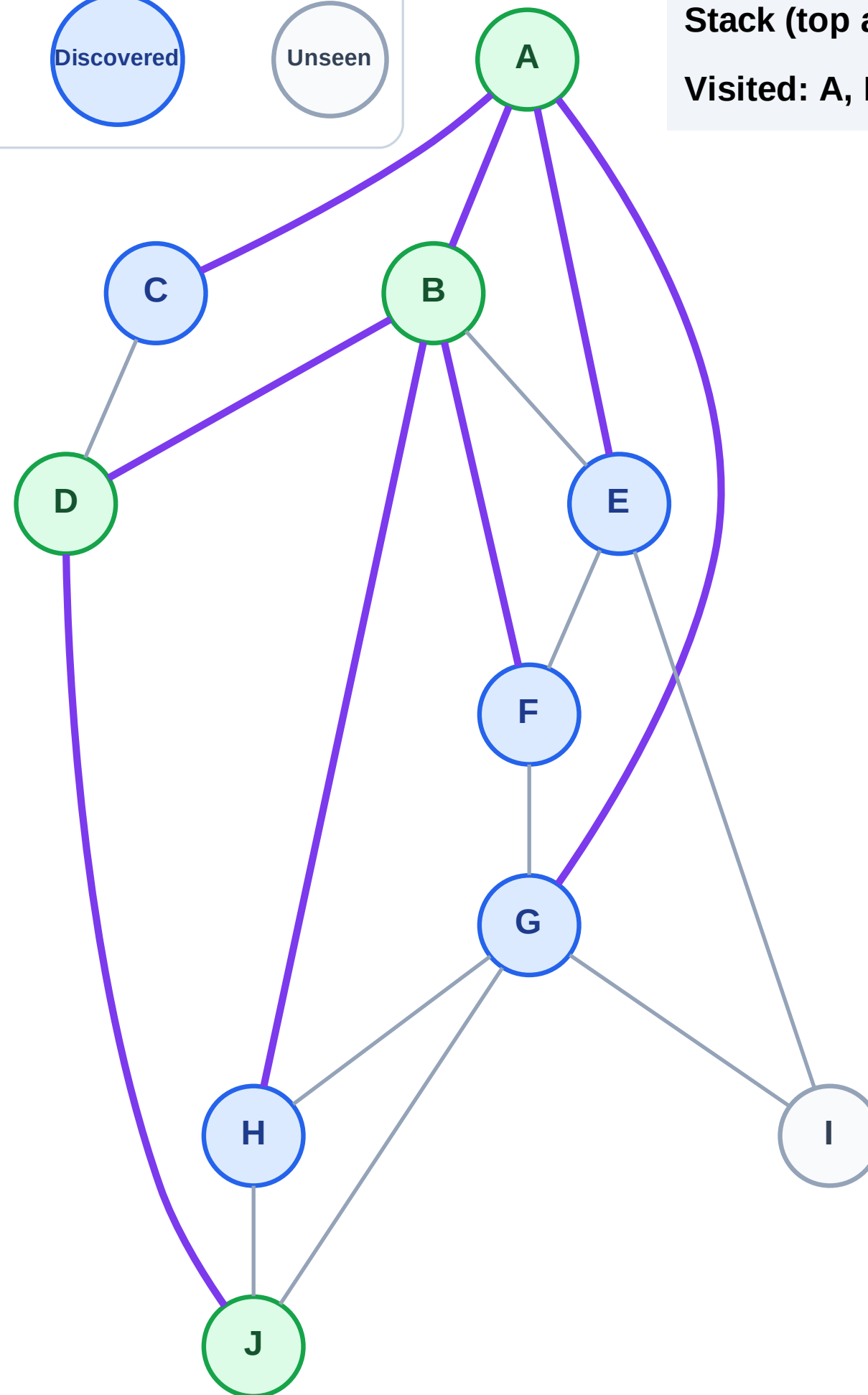
Step 9: No new neighbors from J

Legend



Stack (top at right): G | E | C | H | F

Visited: A, B, D, J



DFS Traversal

Step 10: Process node F

Legend

Complete

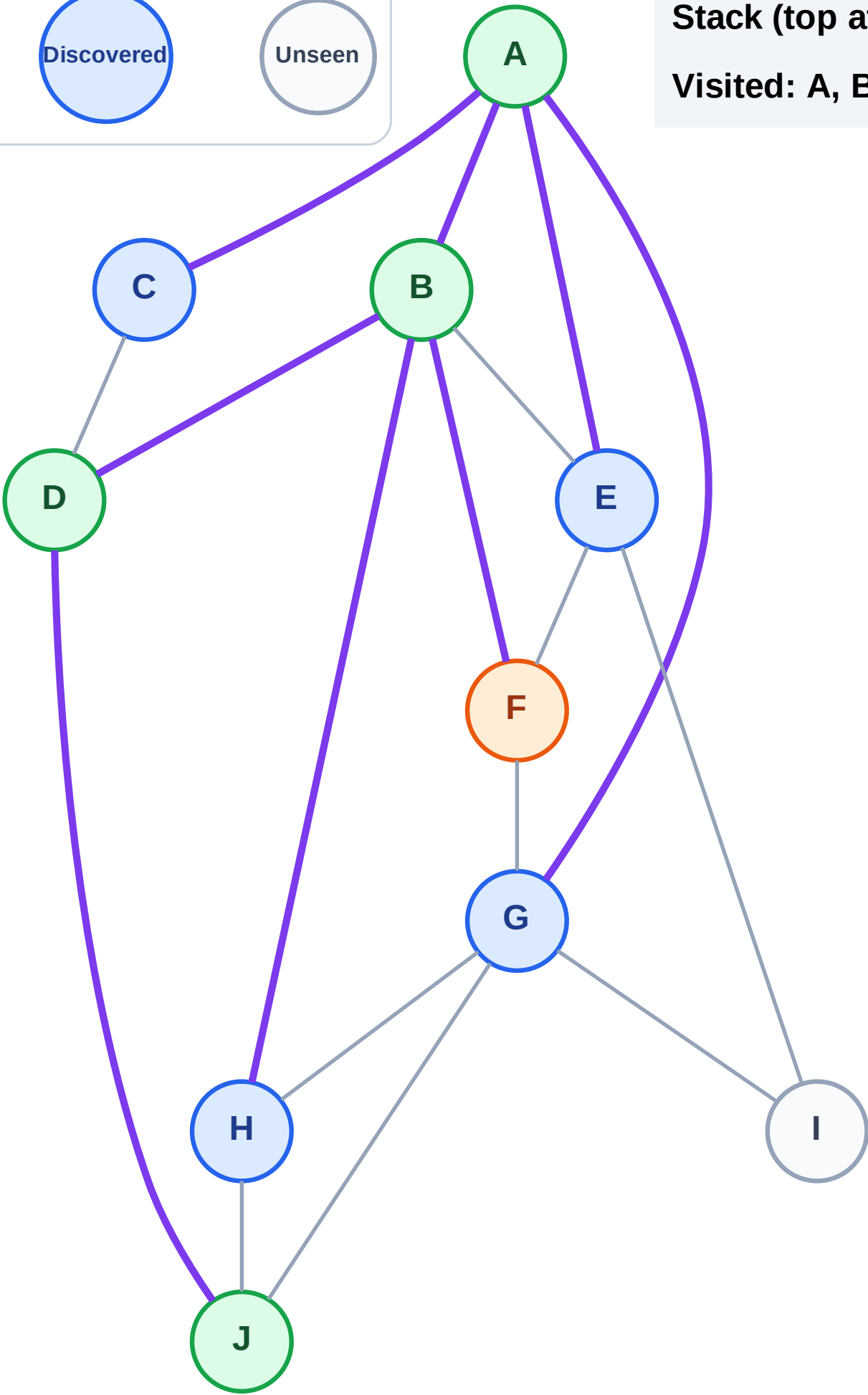
Processing

Discovered

Unseen

Stack (top at right): G | E | C | H

Visited: A, B, D, J



DFS Traversal

Step 11: No new neighbors from F

Legend

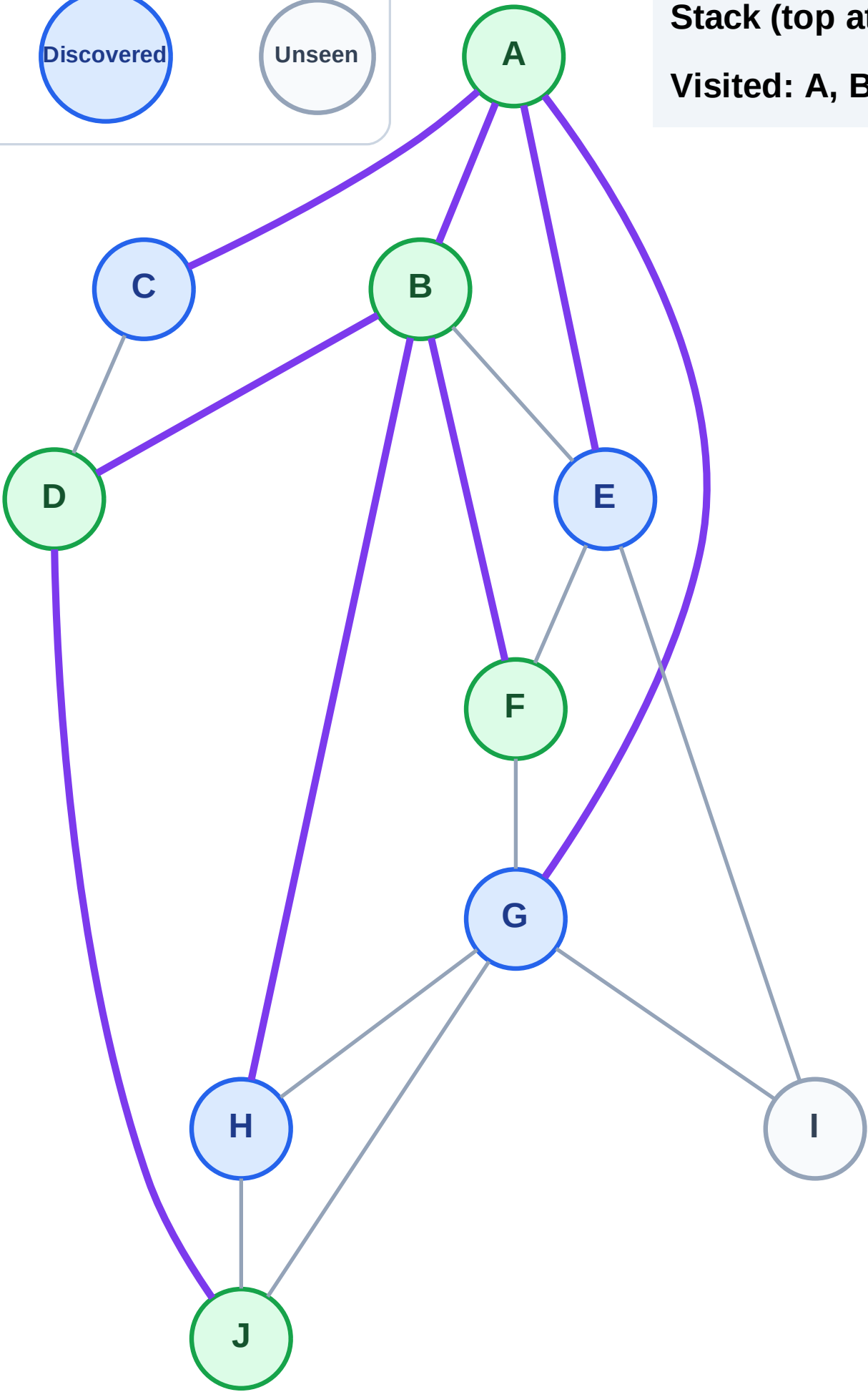
Complete

Processing

Discovered

Unseen

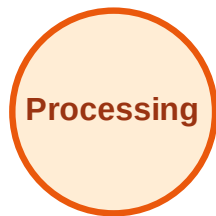
Stack (top at right): G | E | C | H
Visited: A, B, D, F, J



DFS Traversal

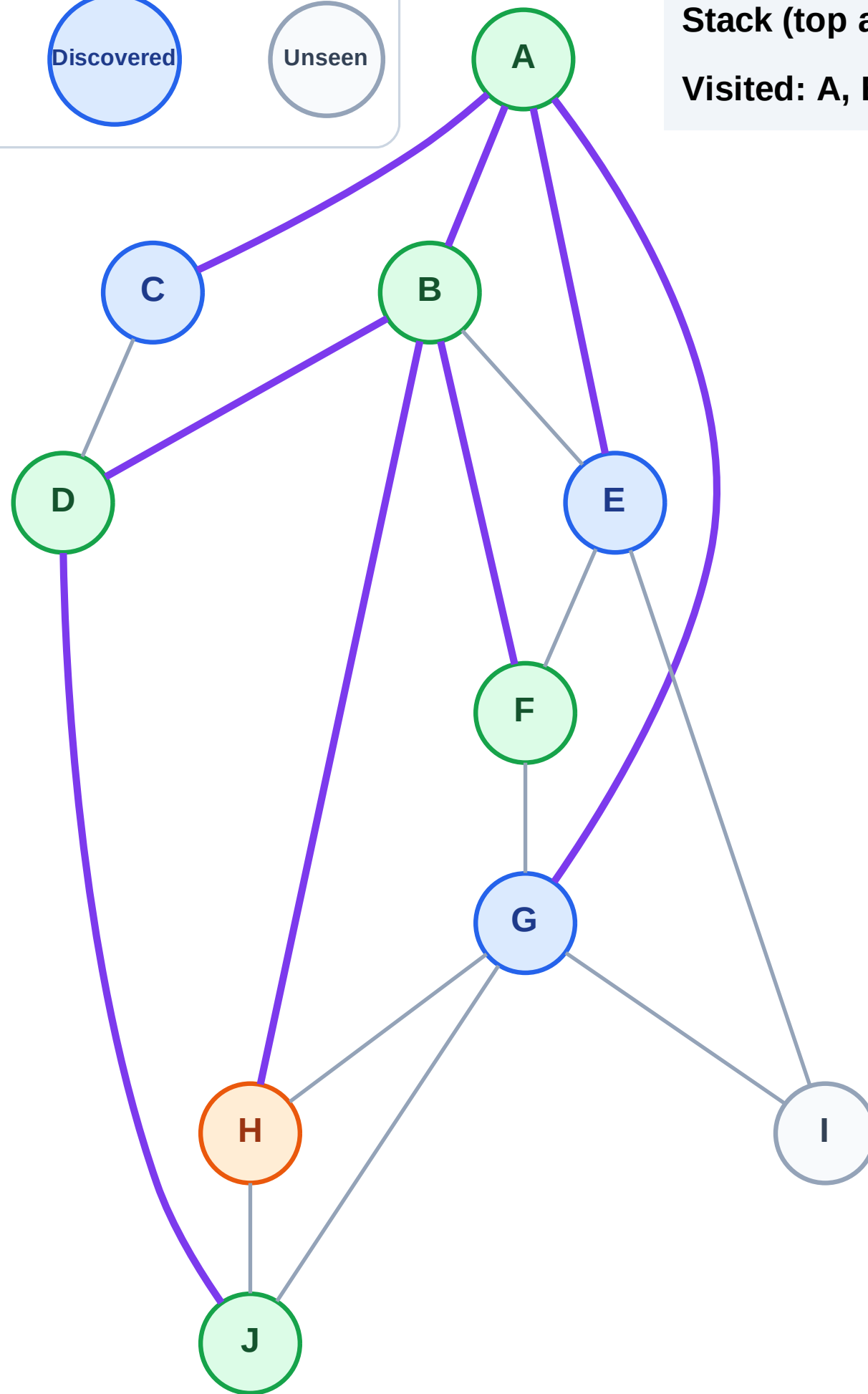
Step 12: Process node H

Legend



Stack (top at right): G | E | C

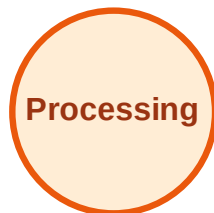
Visited: A, B, D, F, J



DFS Traversal

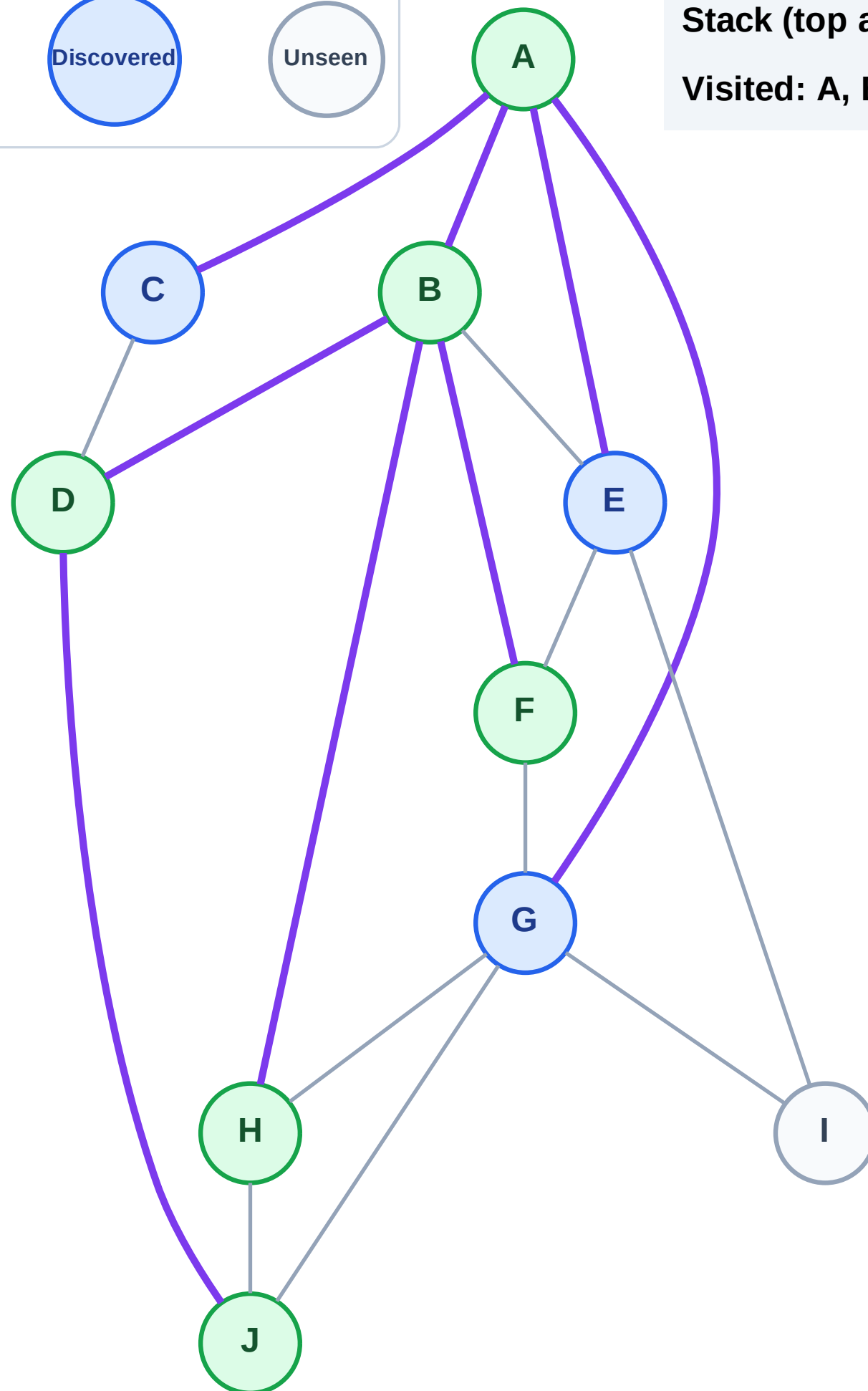
Step 13: No new neighbors from H

Legend



Stack (top at right): G | E | C

Visited: A, B, D, F, H, J



DFS Traversal

Step 14: Process node C

Legend

Complete

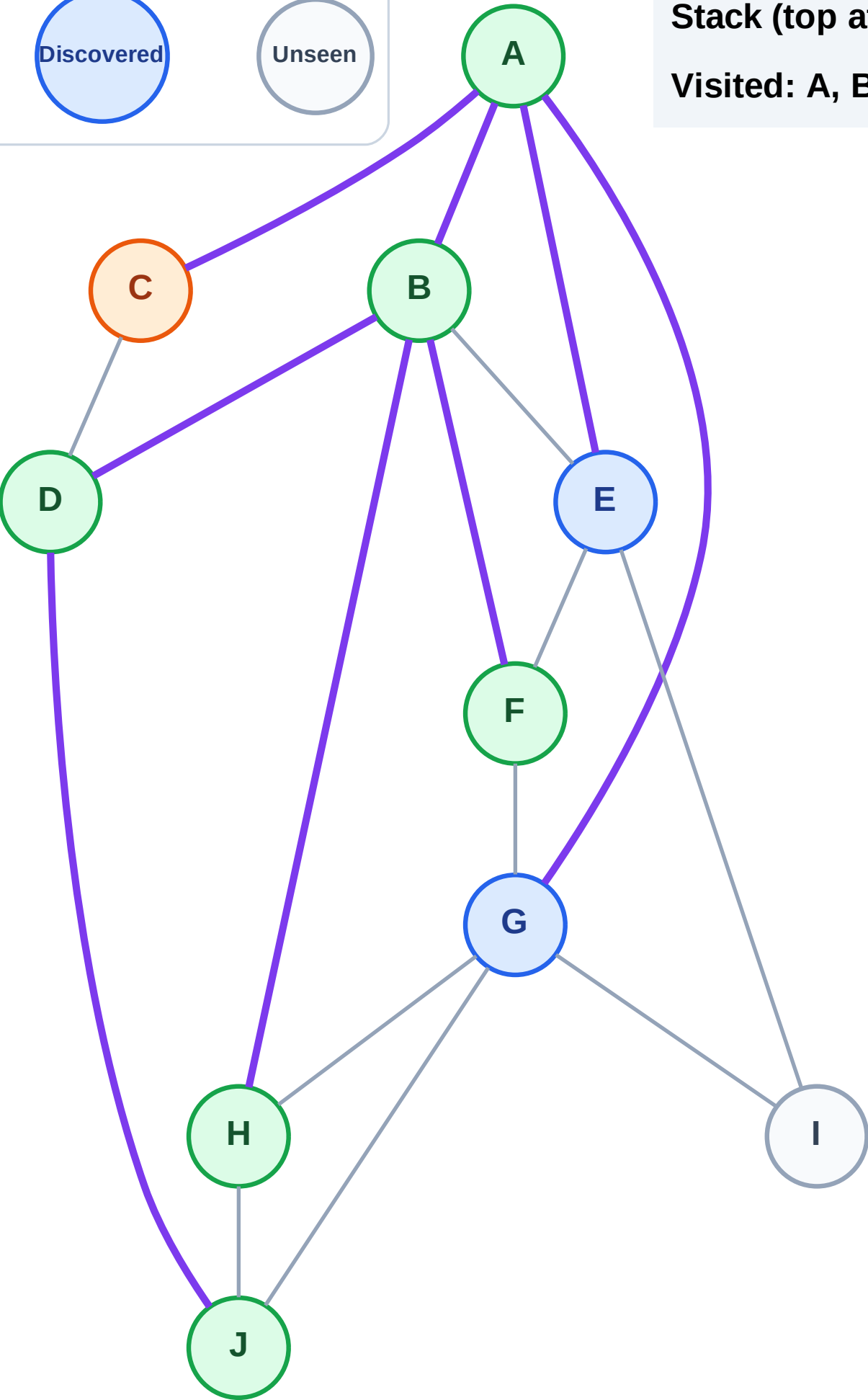
Processing

Discovered

Unseen

Stack (top at right): G | E

Visited: A, B, D, F, H, J



DFS Traversal

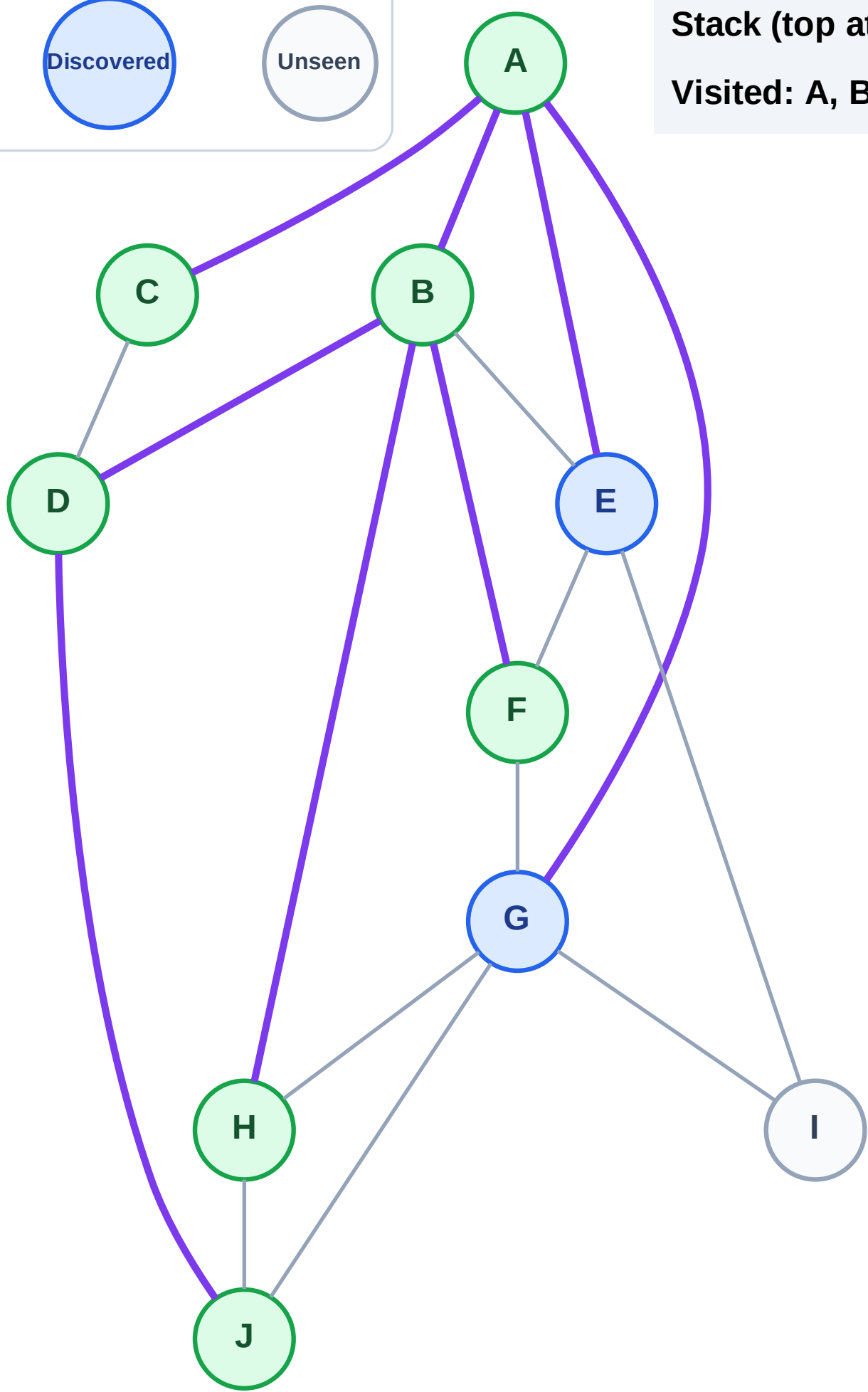
Step 15: No new neighbors from C

Legend



Stack (top at right): G | E

Visited: A, B, C, D, F, H, J



DFS Traversal

Step 16: Process node E

Legend

Complete

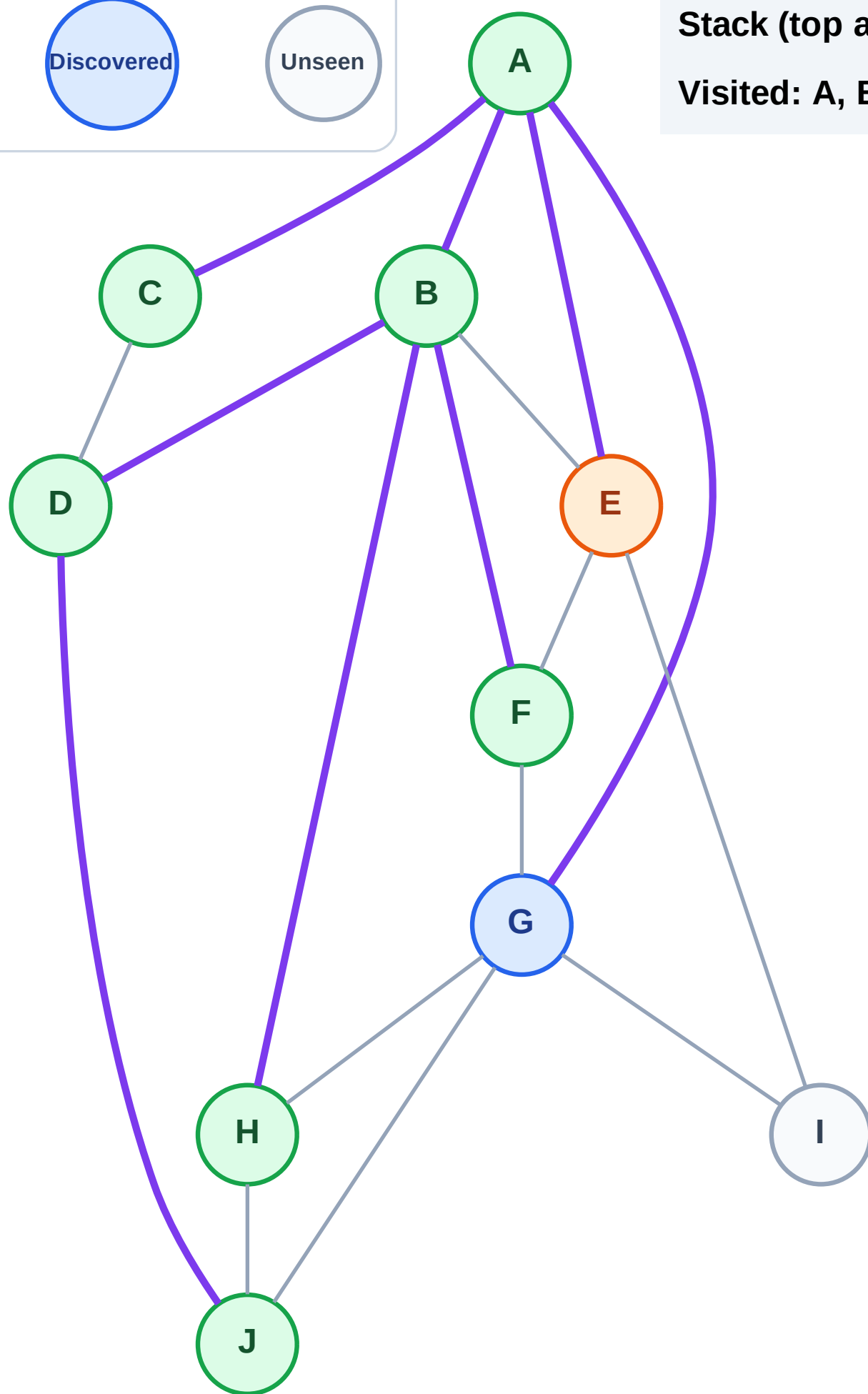
Processing

Discovered

Unseen

Stack (top at right): G

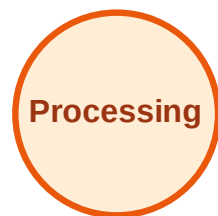
Visited: A, B, C, D, F, H, J



DFS Traversal

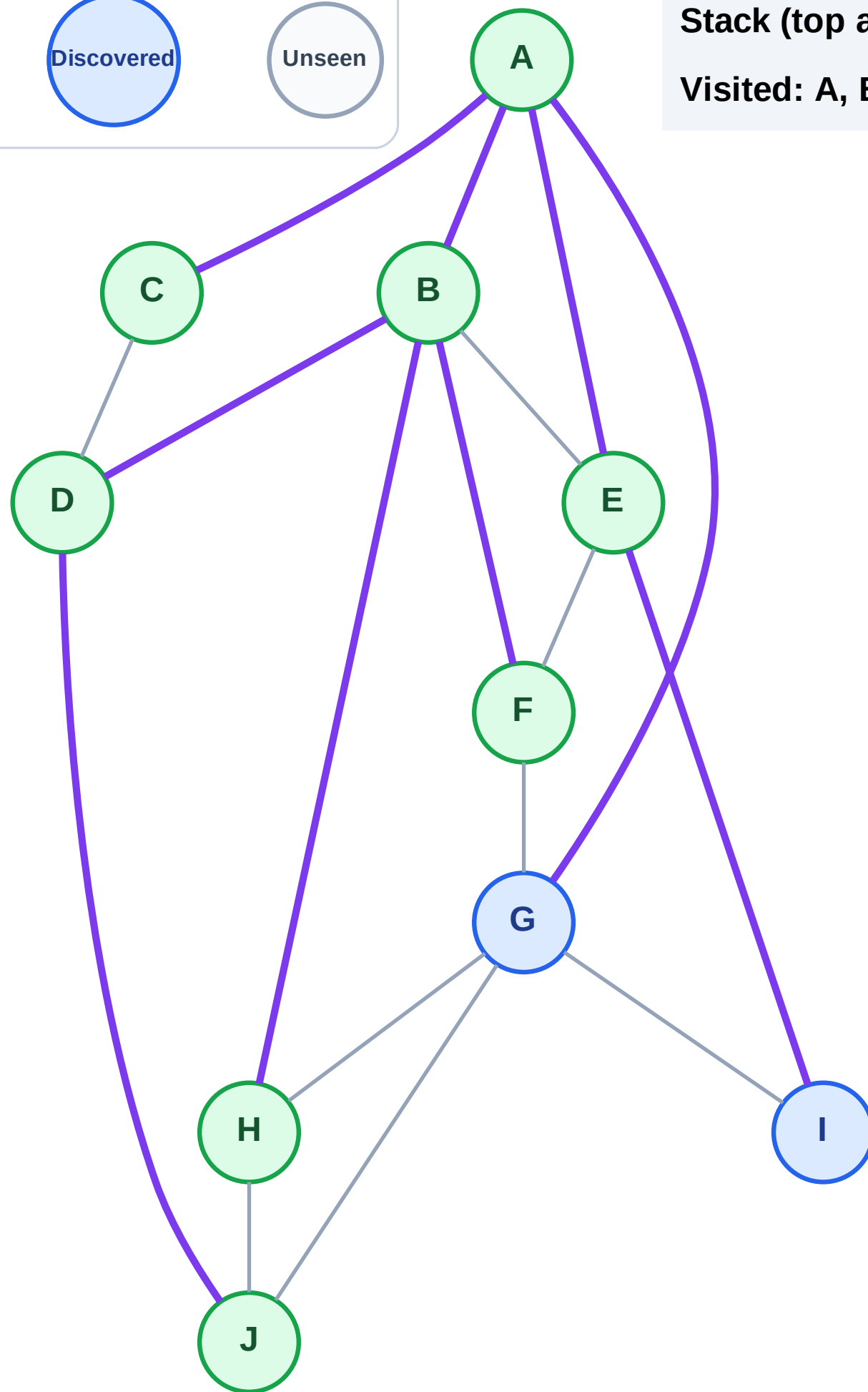
Step 17: Discover I from E

Legend



Stack (top at right): G | I

Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node I

Legend

Complete

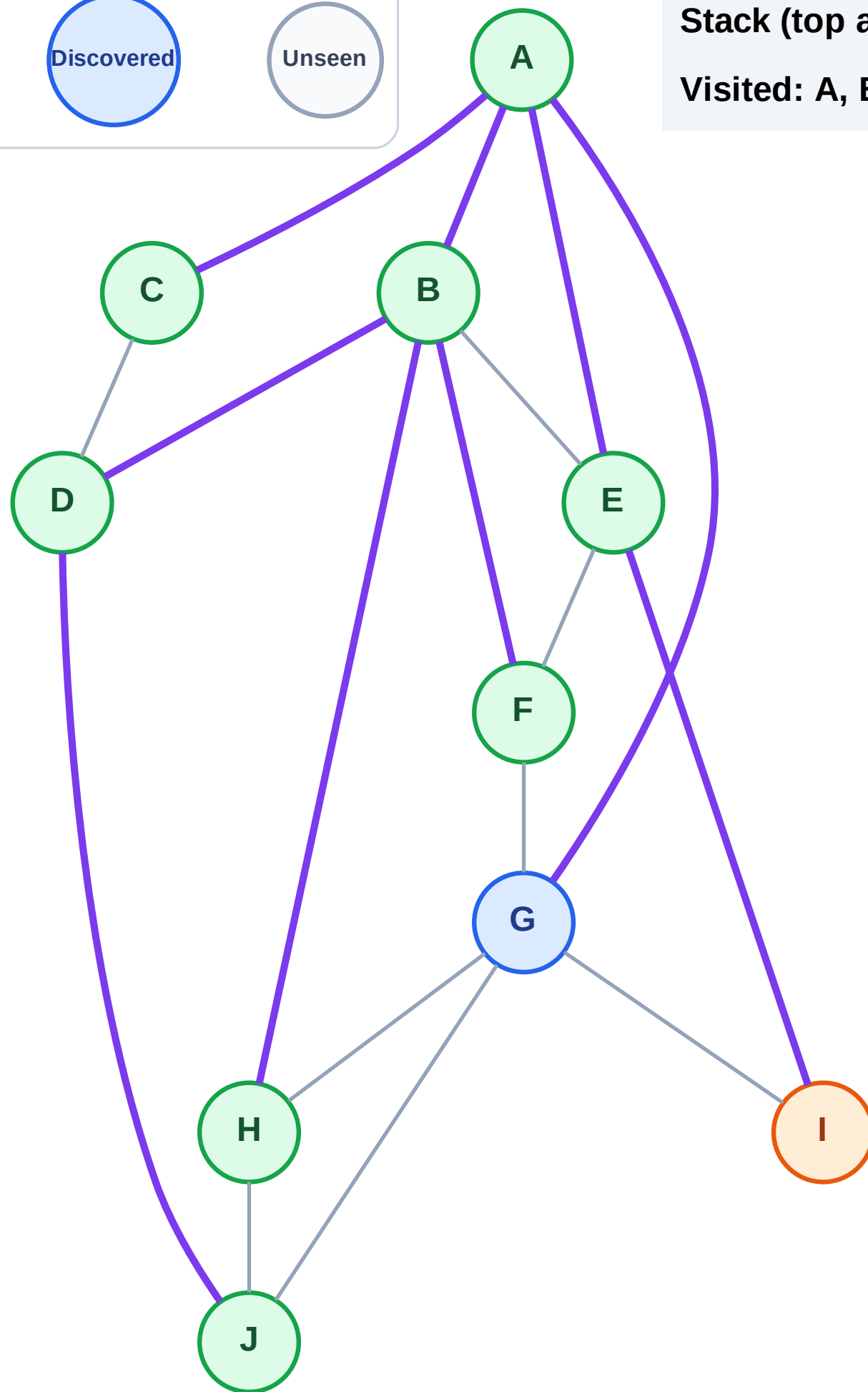
Processing

Discovered

Unseen

Stack (top at right): G

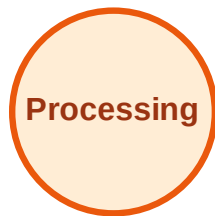
Visited: A, B, C, D, E, F, H, J



DFS Traversal

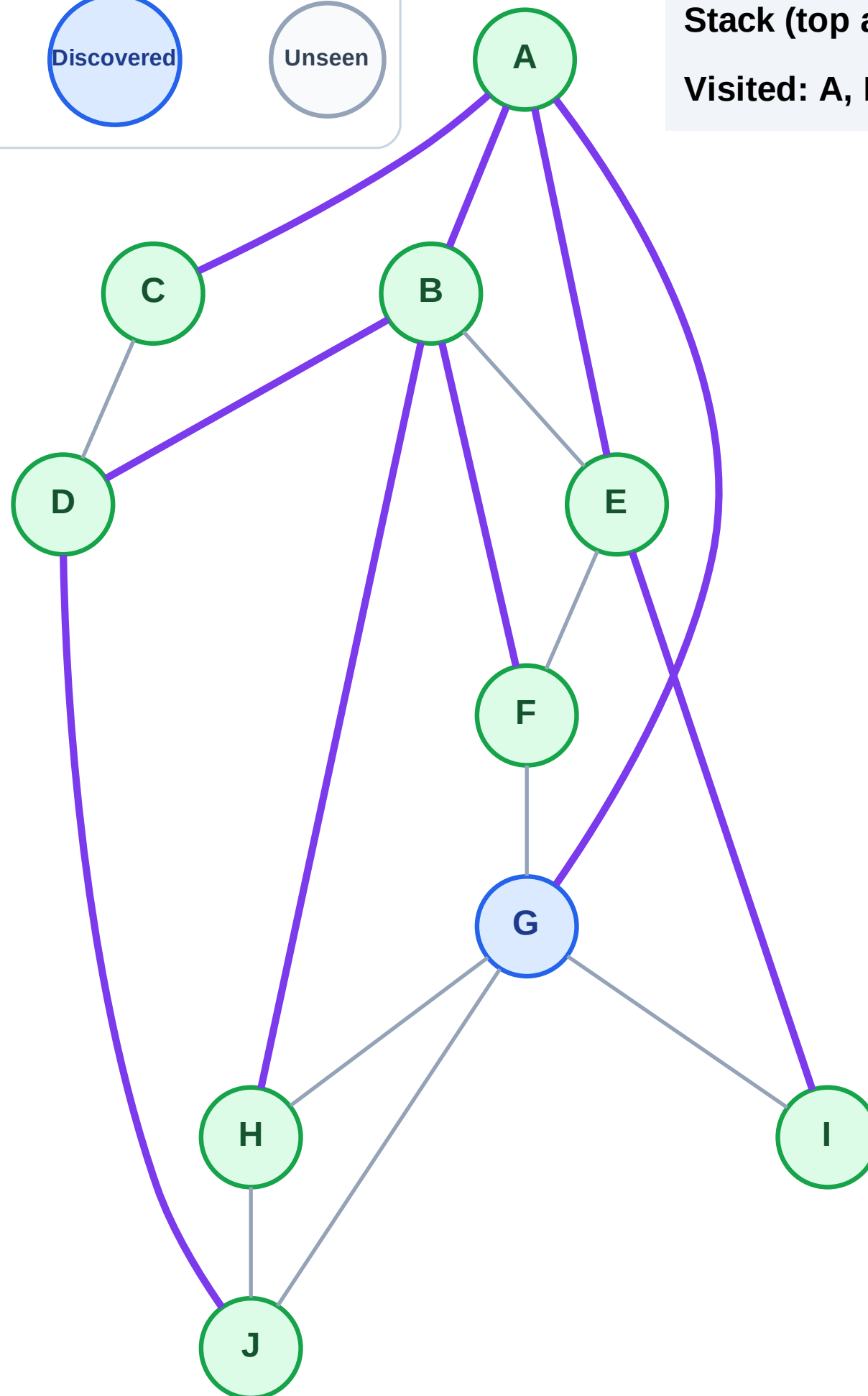
Step 19: No new neighbors from I

Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 20: Process node G

Legend

Complete

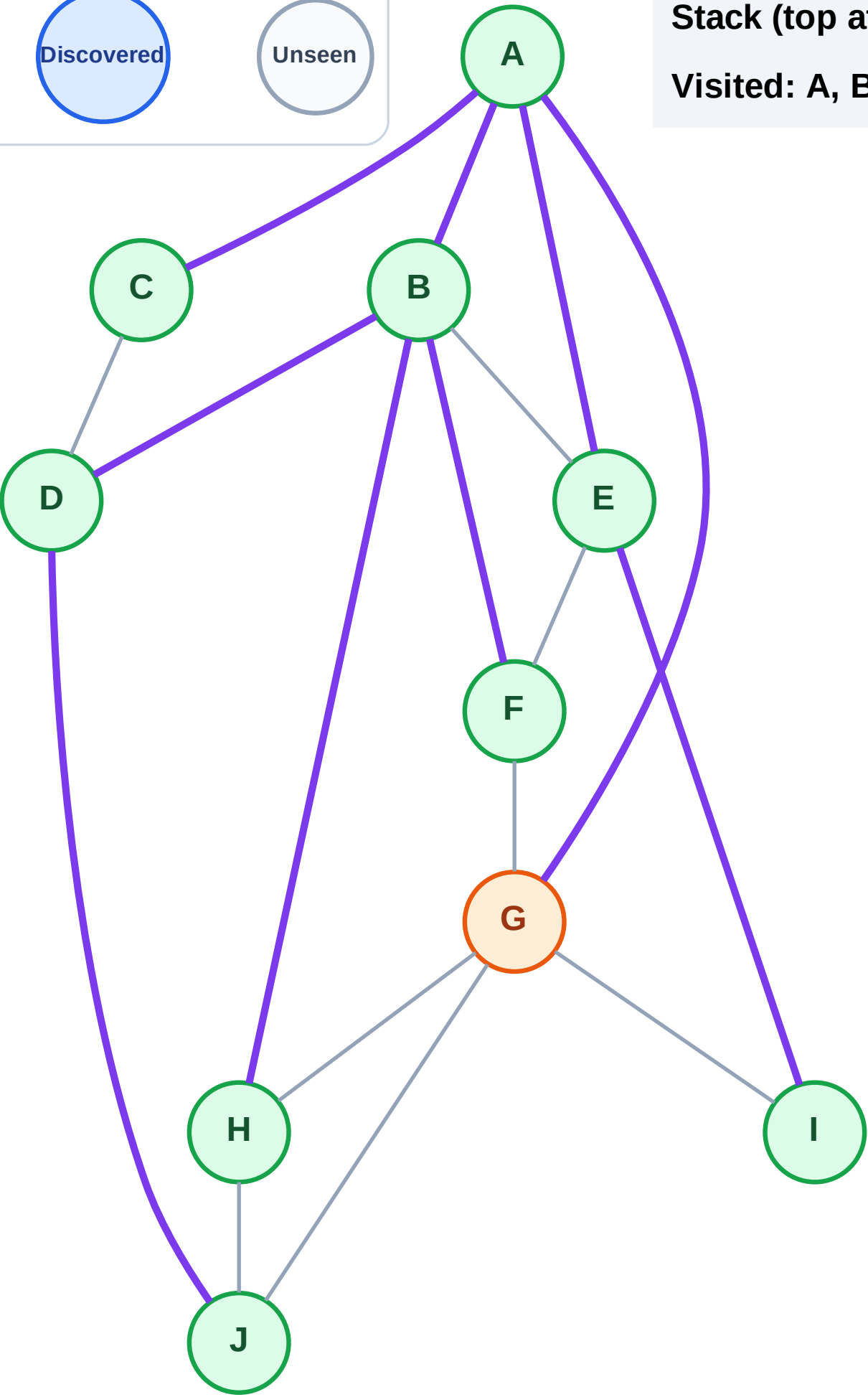
Processing

Discovered

Unseen

Stack (top at right): (empty)

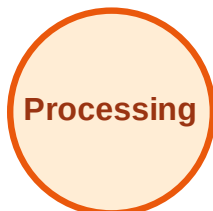
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

